TYPED ROOTED TREES REALIZE THE MULTIPLE PRE-LIE OPERAD

Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

Bruned, Hairer and Zambotti used in [3, 5] typed trees in an essential way to give a systematic description of a canonical renormalisation procedure of stochastic PDEs. Typed trees are rooted trees in which edges are decorated by elements of a fixed set $\mathcal{T}$ of types. They also appear in a context of low dimension topology in [20] (there, described as nested parentheses) and for the description of combinatorial species in [1]. We here study several algebraic structures on these trees, generalizing results of Connes and Kreimer [10], Chapoton and Livernet [9], Grossman and Larson [14], Calaque, Ebrahimi-Fard and Manchon [7].

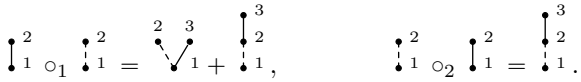
In the work of Bruned, Hairer and Zambotti, the considered trees are typed, with a finite set of types denoted by $\mathfrak{L}$, and labeled. We here forget about the labels and study the algebraic structures induced by types. We first define grafting products of trees, similar to the pre-Lie product of [8]. For any type t, we obtain a pre-Lie product $\bullet_t$ on the space $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$ of $\mathcal{T}$-typed trees whose vertices are decorated by elements of a set $\mathcal{D}$. For example, if $\mid$ and $\vdots$ are two types, if $a, b, c \in \mathcal{D}$, then



Then $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$, equipped with all these products, is a $\mathcal{T}$-multiple pre-Lie algebra (Definition 3.1), also called matching pre-Lie algebras in [25]: for any types t and t', for any $x, y, z \in \mathfrak{g}_{\mathcal{D},\mathcal{T}}$,

$$x \bullet_{t'} (y \bullet_t z) - (x \bullet_{t'} y) \bullet_t z = x \bullet_t (z \bullet_{t'} y) - (x \bullet_t z) \bullet_{t'} y.$$

This relation appears in [4]. We prove in Corollary 3.11 that it is the free $\mathcal{T}$-multiple pre-Lie algebra generated by $\mathcal{D}$, generalizing the result of [9], first mentioned in [4, Proposition 4.21]. Consequently, we obtain a combinatorial description of the operad of $\mathcal{T}$-multiple pre-Lie algebras in terms of $\mathcal{T}$-typed trees with indexed vertices (Theorem 3.17): for example,

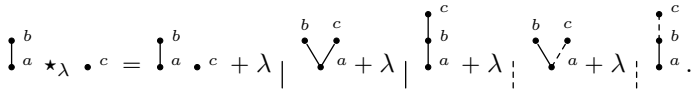


We also give a desription of the Koszul dual operad and of its free algebras in Propositions 3.19 and 3.20, generalizing a result of [8].

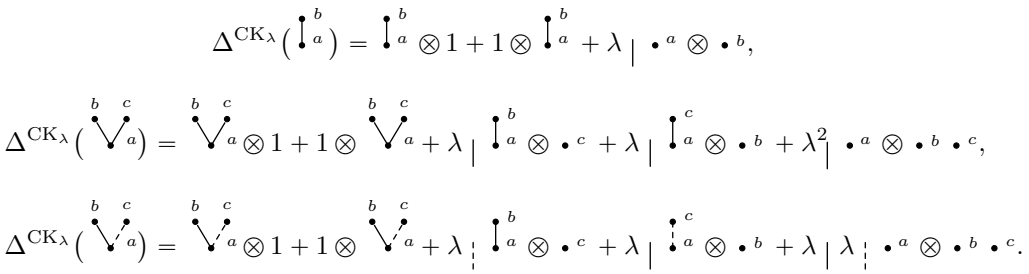
For any family $\lambda = (\lambda_t)_{t\in\mathcal{T}}$ with a finite support, the product $\bullet_\lambda = \sum \lambda_t \bullet_t$ is pre-Lie: using the Guin–Oudom construction [21, 22], we obtain a Hopf algebraic structure $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}} = (S(\mathfrak{g}_{\mathcal{D},\mathcal{T}}), \star_\lambda, \Delta)$ on the symmetric algebra generated by $\mathcal{T}$-typed and $\mathcal{D}$-decorated trees, that is to say on the space of $\mathcal{T}$-typed and $\mathcal{D}$-decorated

Date: doi:10.3842/SIGMA.2021.086; CC BY 4.0.

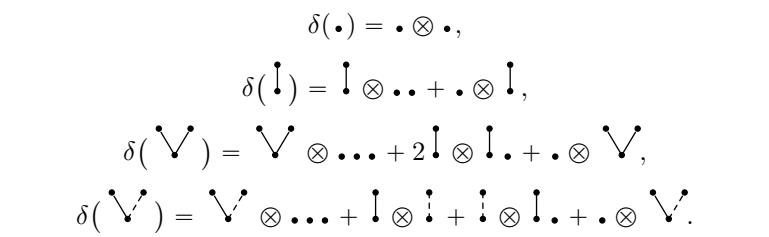
forests. The coproduct Δ is given by partitions of forests into two forests and the $\star_\lambda$ product is given by grafting. For example,



In the non-typed case, we get back the Grossman–Larson Hopf algebra of trees [14]. Dually, we obtain Hopf algebras $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$, generalizing the Connes–Kreimer Hopf algebra [10] of rooted trees. For example,



This Hopf algebra satisfies a universal property in Hochschild cohomology, as does the Connes–Kreimer's Hopf algebra. We describe it in the simpler case where $\mathcal{T}$ is finite (Theorem 5.5). We finally give a second coproduct δ on $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$, such that $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ is a Hopf algebra in the category of $(S(\mathfrak{g}_{\mathcal{D},\mathcal{T}}), m, \delta)$-right comodules, generalizing the result of [7]. This coproduct δ is given by a contraction-extraction process. For example, in the non-decorated case,



We are also interested in morphisms between these objects. We prove that if λ and μ are both nonzero, then the pre-Lie algebras $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$ and $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\mu)$ are isomorphic (Corollary 4.8). Consequently, if λ and μ are both nonzero, the Hopf algebras $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{GL}_\mu}_{\mathcal{D},\mathcal{T}}$ are isomorphic; dually, the Hopf algebras $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{CK}_\mu}_{\mathcal{D},\mathcal{T}}$ are isomorphic (Corollary 5.7). Using Livernet's rigidity theorem [16] and a nonassociative permutative coproduct defined in Proposition 4.1, we prove that if $\lambda \neq 0$, then $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$ is, as a pre-Lie algebra, freely generated by a family of typed trees $\mathcal{D}' = \mathcal{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$ satisfying a condition on the type of edges born from the root (Corollary 4.2). As a consequence, the Hopf algebra $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ of typed and decorated trees is isomorphic to a Connes–Kreimer Hopf algebra of non typed and decorated trees $\mathcal{H}^{\mathrm{CK}}_{\mathcal{D}'}$, and an explicit isomorphism is described with the help of contraction in Proposition 5.9.

This paper is organized as follows: the first section gives the basic definition of typed rooted trees and enumeration results, when the number of types and decorations are finite. The second section is about the $\mathcal{T}$-multiple pre-Lie algebra structures on these trees and the underlying operads. The freeness of the pre-Lie structures on typed decorated trees and its consequences are studied in the third section. In the last section, the dual Hopf algebras $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ are defined, studied and related to the constructions of Bruned, Hairer and Zambotti

[3, 5]: forgetting the labels, the two coproducts they use on a family of typed and partially decorated trees are a subquotient of a the construction presented here.

Since the first version of this paper, written in 2018, multiple pre-Lie algebras have appeared in [4, 6] in a context of SPDEs. When the set of types has more structures (for example a product making it a commutative semigroup), then other possibilities for grafting products of typed trees are studied in [13, 18, 25], giving other interesting structures on these objects.

Notation 1.1.

- We denote by $\mathbb{K}$ a commutative field of characteristic zero. All the objects (vector spaces, algebras, coalgebras, pre-Lie algebras, ...) in this text will be taken over $\mathbb{K}$.
- For any $n \in \mathbb{N}$, we denote by $[n]$ the set $\{1, \dots, n\}$.
- For any set $\mathcal{T}$, we denote by $\mathbb{K}^{\mathcal{T}}$ the set of family $\lambda = (\lambda_t)_{t\in\mathcal{T}}$ of elements of $\mathbb{K}$ indexed by $\mathcal{T}$, and we denote by $\mathbb{K}^{(\mathcal{T})}$ the set of elements $\lambda \in \mathbb{K}^{\mathcal{T}}$ with a finite support. Note that if $\mathcal{T}$ is finite, then $\mathbb{K}^{\mathcal{T}} = \mathbb{K}^{(\mathcal{T})}$.

2. Typed decorated trees

2.1. Definition.

Definition 2.1. Let $\mathcal{D}$ and $\mathcal{T}$ be two nonempty sets.

(1) A $\mathcal{D}$-decorated $\mathcal{T}$-typed forest is a triple $(F, \mathrm{dec}, \mathrm{type})$, where
   - F is a rooted forest. The set of its vertices is denoted by $V(F)$ and the set of its edges by $E(F)$.
   - $\mathrm{dec}\colon V(F) \longrightarrow \mathcal{D}$ is a map.
   - $\mathrm{type}\colon E(F) \longrightarrow \mathcal{T}$ is a map.

   If the underlying rooted forest of F is connected, we shall say that F is a $\mathcal{D}$-decorated $\mathcal{T}$-typed tree.

(2) If $(F, \mathrm{dec}_F, \mathrm{type}_F)$ and $(G, \mathrm{dec}_G, \mathrm{type}_G)$ are two $\mathcal{D}$-decorated $\mathcal{T}$-typed forests, they are isomorphic if there exists a rooted forest isomorphism f from F to G such that for any vertex v of F, $\mathrm{dec}_G(f(v)) = \mathrm{dec}_F(v)$ and for any edge e of F, $\mathrm{type}_G(f(e)) = \mathrm{type}_F(e)$

(3) For any finite set A, we denote by $\mathbb{T}_{\mathcal{T}}(A)$ the set of A-decorated $\mathcal{T}$-typed trees T such that $V(T) = A$ and $\mathrm{dec} = \mathrm{Id}_A$, and by $\mathbb{F}_{\mathcal{T}}(A)$ the set of A-decorated $\mathcal{T}$-typed forests F such that $V(F) = A$ and $\mathrm{dec} = \mathrm{Id}_A$.

(4) For any $n \geqslant 0$, we denote by $\mathbb{T}_{\mathcal{D},\mathcal{T}}(n)$ the set of isomorphism classes of $\mathcal{D}$-decorated $\mathcal{T}$-typed trees T such that $|V(T)| = n$ and by $\mathbb{F}_{\mathcal{D},\mathcal{T}}(n)$ the set of isomorphism classes of $\mathcal{D}$-decorated $\mathcal{T}$-typed forests F such that $|V(F)| = n$. We also put

$$\mathbb{T}_{\mathcal{D},\mathcal{T}} = \bigsqcup_{n\geqslant 0} \mathbb{T}_{\mathcal{D},\mathcal{T}}(n), \qquad \mathbb{F}_{\mathcal{D},\mathcal{T}} = \bigsqcup_{n\geqslant 0} \mathbb{F}_{\mathcal{D},\mathcal{T}}(n).$$

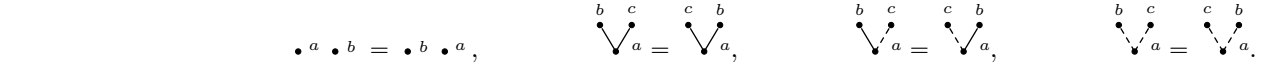
Example 2.2. We shall represent the decorations of the vertices by letters alongside them. If $\mathcal{T}$ contains two elements, represented by $\mid$ and $\vdots$, then

$$\mathbb{F}_{\mathcal{D},\mathcal{T}}(1) = \{\bullet^{\,d}, d \in \mathcal{D}\},$$

$$\mathbb{F}_{\mathcal{D},\mathcal{T}}(2) = \left\{ \bullet^{\,a}\ \bullet^{\,b},\ \begin{smallmatrix} b \\ \mid \\ a \end{smallmatrix},\ \begin{smallmatrix} b \\ \vdots \\ a \end{smallmatrix}, a, b \in \mathcal{D} \right\},$$

$$\mathbb{F}_{\mathcal{D},\mathcal{T}}(3) = \left\{ \bullet^{\,a}\ \bullet^{\,b}\ \bullet^{\,c},\ \begin{smallmatrix} b \\ \mid \\ a \end{smallmatrix}\ \bullet^{\,c},\ \begin{smallmatrix} b \\ \vdots \\ a \end{smallmatrix}\ \bullet^{\,c},\ \begin{smallmatrix} b & c \\ & \diagdown \\ & a \end{smallmatrix},\ \begin{smallmatrix} b & c \\ \diagup & \\ a & \end{smallmatrix},\ \begin{smallmatrix} b & c \\ \diagup & \diagdown \\ a & \end{smallmatrix},\ \begin{smallmatrix} c \\ \mid \\ b \\ \mid \\ a \end{smallmatrix},\ \begin{smallmatrix} c \\ \mid \\ b \\ \vdots \\ a \end{smallmatrix},\ \begin{smallmatrix} c \\ \vdots \\ b \\ \mid \\ a \end{smallmatrix},\ \begin{smallmatrix} c \\ \vdots \\ b \\ \vdots \\ a \end{smallmatrix}, a, b, c \in \mathcal{D} \right\}.$$

Note that for any a, b, $c \in \mathcal{D}$,

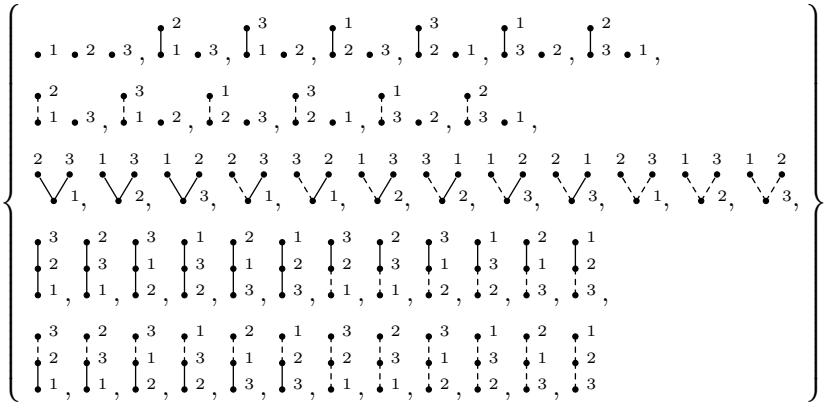


Moreover,

$$\mathbb{F}_{\mathcal{D},\mathcal{T}}([1]) = \{\bullet^{\,1}\},$$



$\mathbb{F}_{\mathcal{D},\mathcal{T}}([3]) =$



Remark 2.3. If $|\mathcal{T}| = 1$, all the edges of elements of $\mathbb{F}_{\mathcal{D},\mathcal{T}}$ have the same type: we work with $\mathcal{D}$-decorated rooted forests. In this case, we shall omit $\mathcal{T}$ in the indices describing the forests, trees, spaces we are considering.

2.2. **Enumeration.** We assume here that $\mathcal{D}$ and $\mathcal{T}$ are finite, of respective cardinality D and T. For all $n \geqslant 0$, we put

$$t_{D,T}(n) = |\mathbb{T}_{\mathcal{T},\mathcal{D}}(n)|, \qquad f_{D,T}(n) = |\mathbb{F}_{\mathcal{T},\mathcal{D}}(n)|,$$
$$T_{D,T}(X) = \sum_{n=1}^{\infty} t_{D,T}(n) X^n, \qquad F_{D,T}(X) = \sum_{n=0}^{\infty} f_{D,T}(n) X^n.$$

As any element of $\mathbb{F}_{\mathcal{T},\mathcal{D}}$ can be uniquely decomposed as the disjoint union of its connected components, which are elements of $\mathbb{T}_{\mathcal{D},\mathcal{T}}$, we obtain

$$F_{D,T}(X) = \prod_{n=1}^{\infty} \frac{1}{(1-X^n)^{t_{D,T}(n)}}. \tag{2.1}$$

We put $\mathcal{T} = \{t_1, \dots, t_T\}$. For any $d \in \mathcal{D}$, we consider

$$B_d \colon \begin{cases} (\mathbb{F}_{\mathcal{D},\mathcal{T}})^T \longrightarrow \mathbb{T}_{\mathcal{D},\mathcal{T}}, \\ (F_1, \dots, F_T) \longmapsto B_d(F_1, \dots, F_T), \end{cases}$$

where $B_d(F_1, \dots, F_T)$ is the tree obtained by grafting the forests $F_1, \dots, F_n$ on a common root decorated by d; the edges from this root to the roots of F_i are of type t_i for any $1 \leqslant i \leqslant T$. Then B_d is injective, homogeneous of degree 1, and moreover $\mathbb{T}_{\mathcal{D},\mathcal{T}}$ is the disjoint union of the $B_d\big((\mathbb{F}_{\mathcal{D},\mathcal{T}})^T\big)$, $d \in \mathcal{D}$. Hence,

$$T_{D,T}(X) = DX(F_{D,T})^T = DX \prod_{n=1}^{\infty} \frac{1}{(1-X^n)^{t_{D,T}(n)T}}. \tag{2.2}$$

Note that (2.2) allows to compute $t_{D,T}(n)$ by induction on n, and (2.1) allows to deduce $f_{D,T}(n)$.

Lemma 2.4. *For any $n \in \mathbb{N}$,*

$$t_{D,T}(n) = \frac{t_{TD,1}(n)}{T}.$$

Proof. By induction on n. If $n = 1$, $t_{D,T}(1) = D$ and $t_{TD,1}(1) = TD$, which gives the result. Let us assume the result at all ranks $k < n$. Then $t_{D,T}(n)T$ is the coefficient of X^n in

$$TDX \prod_{k=1}^{n-1} \frac{1}{(1-X^k)^{t_{D,T}(k)T}} = TDX \prod_{k=1}^{n-1} \frac{1}{(1-X^k)^{t_{TD,1}(k)}},$$

which is precisely $t_{TD,1}(n)$. □

Example 2.5. We obtain

$$\begin{aligned}
t_{D,T}(1) &= D,\\
t_{D,T}(2) &= D^2T,\\
t_{D,T}(3) &= \frac{D^2T(3DT+1)}{2},\\
t_{D,T}(4) &= \frac{D^2T\big(8D^2T^2+3DT+1\big)}{3},\\
t_{D,T}(5) &= \frac{D^2T\big(125D^3T^3+54D^2T^2+31DT+6\big)}{24},\\
t_{D,T}(6) &= \frac{D^2T\big(162D^4T^4+80D^3T^3+45D^2T^2+10DT+3\big)}{15},\\
t_{D,T}(7) &= \frac{D^2T\big(16807D^5T^5+9375D^4T^4+5395D^3T^3+2025D^2T^2+838DT+120\big)}{720}.
\end{aligned}$$

We shall give tables of values of $t_{D,T}(k)$ in the appendix.

3. Multiple pre-Lie algebras

We here fix a nonempty set $\mathcal{T}$ of edges types.

3.1. **Definition.**

Definition 3.1. A $\mathcal{T}$-multiple pre-Lie algebra is a family $(V, (\bullet_t)_{t\in\mathcal{T}})$, where V is a vector space and for all $t \in \mathcal{T}$, $\bullet_t$ is a bilinear product on V such that

$$\forall t, t' \in \mathcal{T}, \quad \forall x, y, z \in V, \qquad x \bullet_{t'} (y \bullet_t z) - (x \bullet_{t'} y) \bullet_t z = x \bullet_t (z \bullet_{t'} y) - (x \bullet_t z) \bullet_{t'} y.$$

Remark 3.2. For any $t \in \mathcal{T}$, $(V, \bullet_t)$ is a pre-Lie algebra. More generally, for any family $\lambda = (\lambda_t)_{t\in\mathcal{T}} \in \mathbb{K}^{(\mathcal{T})}$, putting $\bullet_\lambda = \sum \lambda_t \bullet_t$, $(V, \bullet_\lambda)$ is a pre-Lie algebra.

Proposition 3.3. *Let $\mathcal{D}$ be any set; we denote by $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$ the vector space generated by $\mathbb{T}_{\mathcal{D},\mathcal{T}}$. For any $T, T' \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$, $v \in V(T)$ and $t \in \mathcal{T}$, we denote by $T \bullet_t^{(v)} T'$ the $\mathcal{D}$-decorated $\mathcal{T}$-typed tree obtained by grafting T' on v (that is to say adding an edge between v and the root of T'), the created edge being of type t. We then define a product $\bullet_t$ on $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$ by*

$$\forall T, T' \in \mathbb{T}_{\mathcal{D},\mathcal{T}}, \qquad T \bullet_t T' = \sum_{v\in V(T)} T \bullet_t^{(v)} T'.$$

Then $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, (\bullet_t)_{t\in\mathcal{T}})$ is a $\mathcal{T}$-multiple pre-Lie algebra.

Proof. Let T, T', T'' be elements of $\mathbb{T}_{\mathcal{D},\mathcal{T}}$ and $t', t'' \in \mathcal{T}$.

$$\begin{aligned}
&(T \bullet_{t'} T') \bullet_{t''} T'' - T \bullet_{t'} (T' \bullet_{t''} T'') \\
&= \sum_{v\in V(T),v'\in V(T)\sqcup V(T')} \left(T \bullet_{t'}^{(v)} T'\right) \bullet_{t''}^{(v')} T'' - \sum_{v\in V(T),v'\in V(T')} T \bullet_{t'}^{(v)} \left(T' \bullet_{t''}^{(v')} T''\right) \\
&= \sum_{v\in V(T),v'\in V(T)} \left(T \bullet_{t'}^{(v)} T'\right) \bullet_{t''}^{(v')} T'' + \sum_{v\in V(T),v'\in V(T')} \left(T \bullet_{t'}^{(v)} T'\right) \bullet_{t''}^{(v')} T'' - T \bullet_{t'}^{(v)} \left(T' \bullet_{t''}^{(v')} T''\right) \\
&= \sum_{v\in V(T),v'\in V(T)} \left(T \bullet_{t'}^{(v)} T'\right) \bullet_{t''}^{(v')} T'' \\
&= \sum_{v\in V(T),v'\in V(T)} \left(T \bullet_{t''}^{(v')} T''\right) \bullet_{t'}^{(v)} T' \\
&= (T \bullet_{t''} T'') \bullet_{t'} T' - T \bullet_{t''} (T'' \bullet_{t'} T').
\end{aligned}$$

So $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$ is indeed a $\mathcal{T}$-multiple pre-Lie algebra. □

Example 3.4. If $a, b, c \in \mathcal{D}$ and $\left|, \right. \vdots \in \mathcal{T}$,



3.2. **Guin–Oudom extension of the pre-Lie products.**

Notation 3.5. Let $(\partial_t)_{t\in\mathcal{T}}$ be a family of formal symbols indexed by $\mathcal{T}$. For any vector space V, we denote

$$V^{\oplus\mathcal{T}} = V \otimes \mathrm{Vect}(\partial_t,\ t \in \mathcal{T}).$$

Then

$$V^{\oplus\mathcal{T}} = \bigoplus_{t\in\mathcal{T}} V \otimes \partial_t.$$

In order to enlighten the notations, we write $v\partial_t$ instead of $v \otimes \partial_t$ for any $v \in V$ and for any $t \in \mathcal{T}$.

Lemma 3.6. *If for any $t \in \mathcal{T}$, $\bullet_t$ is a bilinear product on a vector space V, we define $\bullet : \left(V^{\oplus\mathcal{T}}\right)^{\otimes 2} \longrightarrow V^{\oplus\mathcal{T}}$ by*

$$x\partial_t \bullet x'\partial_{t'} = (x \bullet_{t'} y)\partial_t.$$

Then $(V, (\bullet_t)_{t\in\mathcal{T}})$ is a $\mathcal{T}$-multiple pre-Lie algebra if, and only if, $\left(V^{\oplus\mathcal{T}}, \bullet\right)$ is a pre-Lie algebra.

Proof. Let $x, x', x'' \in V$, $t, t', t'' \in \mathcal{T}$. Then

$$x\partial_t \bullet (x'\partial_{t'} \bullet x''\partial_{t''}) - (x\partial_t \bullet x'\partial_{t'}) \bullet x''\partial_{t''} = \left((x \bullet_{t'} x') \bullet_{t''} x'' - x \bullet_{t'} (x' \bullet_{t''} x'')\right)\partial_t,$$

which implies the result. □

Notation 3.7. The symmetric algebra $S(V)$ is given with its usual coproduct Δ, making it a bialgebra:

$$\forall x \in V, \qquad \Delta(x) = x \otimes 1 + 1 \otimes x.$$

We shall use Sweedler's notation: for any $w \in S(V)$, $\Delta(w) = \sum w^{(1)} \otimes w^{(2)}$. The counit of this coproduct is denoted by ε: this is the unique algebra morphism from $S(V)$ to $\mathbb{K}$ sending any $v \in V$ to 0.

Theorem 3.8. *Let V be a $\mathcal{T}$-multiple pre-Lie algebra. One can define a product*

$$\bullet\colon\ S(V)\otimes S\big(V^{\oplus\mathcal{T}}\big)\longrightarrow S(V)$$

in the following way: for any $u,v\in S(V)$, $w\in S\big(V^{\oplus\mathcal{T}}\big)$, $x\in V$, $t\in\mathcal{T}$,

$$\begin{aligned}1\bullet w&=\varepsilon(w),\\ u\bullet 1&=u,\\ uv\bullet w&=\sum\big(u\bullet w^{(1)}\big)\big(v\bullet w^{(2)}\big),\\ u\bullet w(x\partial_t)&=(u\bullet w)\bullet_t x-x\bullet(w\bullet_t x),\end{aligned}$$

where $\bullet_t$ is extended to $S(V)\otimes V$ and $S\big(V^{\oplus\mathcal{T}}\big)\otimes V$ by the following: for any $x_1,\dots,x_k,x\in V$, for any $t_1,\dots,t_k\in\mathcal{T}$,

$$x_1\cdots x_k\bullet_t x=\sum_{i=1}^{k}x_1\cdots(x_i\bullet_t x)\cdots x_k,$$

$$(x_1\partial_{t_1})\cdots(x_k\partial_{t_k})\bullet_t x=\sum_{i=1}^{k}(x_1\partial_{t_1})\cdots((x_i\bullet_t x)\partial_{t_i})\cdots(x_k\partial_{t_k}).$$

Proof. Uniqueness. The last formula allows to compute $x\bullet w$ for any $x\in V$ and $w\in S\big(V^{\oplus\mathcal{T}}\big)$ by induction on the length of w; the other ones allow to compute $u\bullet w$ for any $u\in S(V)$ by induction on the length on u. So this product $\bullet$ is unique.

Existence. Let us use the Guin–Oudom construction [21, 22] on the pre-Lie algebra $V^{\otimes\mathcal{T}}$. We obtain a product $\bullet$ defined on $S\big(V^{\oplus\mathcal{T}}\big)$ such that for any $u,v,w\in S\big(V^{\oplus\mathcal{T}}\big)$, $x\in V^{\oplus\mathcal{T}}$:

$$\begin{aligned}1\bullet w&=\varepsilon(w),\\ u\bullet 1&=u,\\ uv\bullet w&=\sum\big(u\bullet w^{(1)}\big)\big(v\bullet w^{(2)}\big),\\ u\bullet wx&=(u\bullet w)\bullet x-x\bullet(w\bullet x).\end{aligned}$$

Let $f\colon\mathcal{T}\longrightarrow\mathbb{K}$ be any nonzero map. We consider the surjective algebra morphism $F\colon S\big(V^{\oplus\mathcal{T}}\big)\longrightarrow S(V)$, sending $x\partial_t$ to $f(t)x$ for any $x\in V$, $t\in\mathcal{T}$. Its kernel is generated by the elements $X_{t,t'}x=(f(t')\partial_t-f(t)\partial_{t'})x$, where $x\in V$ and $t,t'\in\mathcal{T}$. We denote by J the vector space generated by the elements $X_{t,t'}x$. Let us prove that for any $w\in S\big(V^{\oplus\mathcal{T}}\big)$, $J\bullet w\subseteq J$ by induction on the length n of w. If $n=0$, we can assume that $w=1$ and this is obvious. If $n=1$, we can assume that $w=x'\partial_{t''}$. Then

$$X_{t,t'}x\bullet w=(f(t')\partial_t-f(t)\partial_{t'})x\bullet_{t''}x'=X_{t,'t't'}x\bullet_{t''}x'\in J.'$$

Let us assume the result at rank $n-1$. We can assume that $w=w'x'\partial_t$, the length of w' being $n-1$. For any $x\in J$,

$$x\bullet w=(x\bullet w')\bullet x'-x\bullet(w'\bullet x').$$

The length of w' and $w'\bullet x'$ is $n-1$, so $x\bullet w'$ and $x\bullet(w'\bullet x')$ belong to J. From the case $n=1$, $(x\bullet w')\bullet x'\in J$, so $x\bullet w\in J$.

For any $x\in J$, $u,v\in S\big(V^{\oplus\mathcal{T}}\big)$,

$$xu\bullet v=\underbrace{\big(x\bullet v^{(1)}\big)}_{\in J}\big(u\bullet v^{(2)}\big)\in\mathrm{Ker}(F).$$

This proves that $\mathrm{Ker}(F)\bullet S\big(V^{\oplus\mathcal{T}}\big)\subseteq\mathrm{Ker}(F)$. Hence, $\bullet$ induces a product also denoted by $\bullet$, defined from $S(V)\otimes S\big(V^{\otimes\mathcal{T}}\big)$ to $S(V)$. It is not difficult to show that it does not depend on the choice of f and satisfies the required properties. □

Definition 3.9. Let $d \in \mathcal{D}$, $T_1, \dots, T_k \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$, $t_1, \dots, t_k \in \mathcal{T}$. We denote by

$$B_d\Big(\prod_{i\in[k]} T_i\partial_{t_i}\Big)$$

the $\mathcal{T}$-typed $\mathcal{D}$-decorated tree obtained by grafting $T_1, \dots, T_k$ on a common root decorated by d, the edge between this root and the root of T_i being of type t_i for any i. This defines a map $B_d \colon S\left(\mathrm{Vect}(\mathbb{T}_{\mathcal{D},\mathcal{T}})^{\oplus\mathcal{T}}\right) \longrightarrow S(\mathrm{Vect}(\mathbb{T}_{\mathcal{D},\mathcal{T}}))$.

Lemma 3.10. *For any $d \in \mathcal{D}$, $T_1, \dots, T_k \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$, $t_1, \dots, t_k \in \mathcal{T}$,*

$$B_d\Big(\prod_{i\in[k]} T_i\partial_{t_i}\Big) = \bullet^d \bullet \prod_{i\in[k]} T_i\partial_{t_i}.$$

Proof. We write $F = \prod_{i\in[k]} T_i\partial_{t_i}$: the integer k is unique and denoted by $\ell(F)$. We proceed by induction on $\ell(F)$. If $\ell(F) = 0$, then $F = 1$ and $\bullet^d \bullet 1 = \bullet^d = B_d(1)$. let us assume the result for any forest F' with $\ell(F') = \ell(F) - 1$. Putting $k = \ell(F)$, we can write $F = F'T\partial_t$, with $\ell(F') = \ell(F) - 1$, $T = T_k$ and $t = t_k$. Then

$$\begin{aligned}
\bullet^d \bullet F &= (\bullet^d \bullet F') \bullet T\partial_t - \bullet^d \bullet (F' \bullet T\partial_t) \\
&= B_d(F') \bullet_t T - B_d(F' \bullet_t T) \\
&= B_d(F'T\partial_t) + B_d(F' \bullet_t T) - B_d(F' \bullet_t T) \\
&= B_d(F).
\end{aligned}$$

So the result holds for all forests. □

Corollary 3.11. *Let A be a $\mathcal{T}$-multiple pre-Lie algebra and, for any $d \in \mathcal{D}$, $a_d \in A$. There exists a unique $\mathcal{T}$-multiple algebra morphism $\phi \colon \mathfrak{g}_{\mathcal{D},\mathcal{T}} \longrightarrow A$, such that for any $d \in \mathcal{D}$, $\phi(\bullet^d) = a_d$. In other words, $\mathfrak{g}_{\mathcal{T},\mathcal{D}}$ is the free $\mathcal{T}$-multiple pre-Lie algebra generated by $\mathcal{D}$.*

Proof. Uniqueness. Using the Guin–Oudom product and lemma 3.10, ϕ is the unique linear map inductively defined by

$$\phi\Big(B_d\Big(\prod_{i\in[k]} T_i\partial_{t_i}\Big)\Big) = a_d \bullet \prod_{i\in[k]} \phi(T_i)\partial_{t_i}.$$

Existence. Let $T, T' \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$ and $t \in \mathcal{T}$. Let us prove that $\phi(T \bullet_t T') = \phi(T) \bullet_t \phi(T')$ by induction on $n = |T|$. If $n = 1$, we assume that $T = \bullet^d$. Then $T \bullet_t T' = B_d(T'\partial_t)$, so

$$\phi(T \bullet_t T') = a_d \bullet (\phi(T')\partial_t = a_d \bullet_t \phi(T') = \phi(T) \bullet_t \phi(T').$$

Let us assume the result at all ranks $< |T|$. We put

$$T = B_d\Big(\prod_{i=1}^{k} T_i\partial_{t_i}\Big).$$

By definition of the pre-Lie product of $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$ in terms of grafting,

$$T \bullet T' = B_d\Big(\prod_{i=1}^{k} T_i \partial_{t_i} T' \partial_t\Big) + \sum_{j=1}^{k} B_d\Big(\prod_{i\neq j} T_i \partial_{t_i} (T_j \bullet_t T') \partial_{t_j}\Big),$$

$$\begin{aligned}
\phi(T \bullet T') &= a_d \bullet \prod_{i=1}^{k} \phi(T_i)\partial_{t_i}\phi(T')\partial_t + \sum_{j=1}^{k} a_d \bullet \prod_{i\neq j} \phi(T_i)\partial_{t_i}(\phi(T_j \bullet_t T'))\partial_{t_j} \\
&= a_d \bullet \prod_{i=1}^{k} \phi(T_i)\partial_{t_i}\phi(T')\partial_t + \sum_{j=1}^{k} a_d \bullet \prod_{i\neq j} \phi(T_i)\partial_{t_i}(\phi(T_j) \bullet_t \phi(T'))\partial_{t_j} \\
&= a_d \bullet \prod_{i=1}^{k} \phi(T_i)\partial_{t_i}\phi(T')\partial_t + a_d \bullet \Big(\Big(\prod_{i=1}^{k} \phi(T_i)\partial_{t_i}\Big) \bullet \phi(T')\partial_t\Big) \\
&= \Big(a_d \bullet \prod_{i=1}^{k} \phi(T_i)\partial_{t_i}\Big) \bullet \phi(T')\partial_t \\
&= \phi(T) \bullet_t \phi(T').
\end{aligned}$$

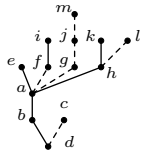
So ϕ is a $\mathcal{T}$-multiple pre-Lie algebra morphism. □

3.3. **Operad of typed trees.** We now describe an operad of typed trees, in the category of species. We refer to [2, 17, 19] for notations and definitions on operads.

Notation 3.12. Let A be a finite set. If $T \in \mathbb{T}_{\mathcal{T}}(A)$ and $a \in T$:

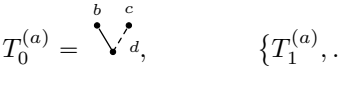
(1) The subtrees formed by the connected components of the set of vertices, descendants of a (a excluded) are denoted by $T_1^{(a)}, \dots, T_{n_a}^{(a)}$. The type of the edge from a to the root of $T_i^{(a)}$ is denoted by t_i.
(2) The tree formed by the vertices of T which are not in $T_1^{(a)}, \dots, T_{n_a}^{(a)}$, at the exception of a, is denoted by $T_0^{(a)}$.

Example 3.13. Let us consider the following tree:

$T =$ 

with $a, \dots, m \in \mathcal{D}$, then

$T_0^{(a)} =$

, $\qquad \big\{T_1^{(a)}, \dots, T_n^{(a)}\big\} =$ 

.

Proposition 3.14. *For any nonempty finite set A, we denote by $\mathcal{P}_{\mathcal{T}}(A)$ the vector space generated by $\mathbb{T}_{\mathcal{T}}(A)$. We define a composition $\circ$ on $\mathcal{P}_{\mathcal{T}}$ in the following way: for any $T \in \mathbb{T}_{\mathcal{T}}(A)$, $T' \in \mathbb{T}_{\mathcal{T}}(B)$ and $a \in A$,*

$$T \circ_a T' = \sum_{v_1,\dots,v_{n_a} \in V(T')} \Big(\cdots\Big(\big(T_0^{(a)} \bullet_{a'}^{(t_0)} T'\big) \bullet_{v_1}^{(t_1)} T_1^{(a)}\Big)\cdots\Big) \bullet_{v_{n_a}}^{(t_{n_a})} T_{n_a}^{(a)},$$

where a' is the direct ascendant of a in T and t_0 is the type of the edge between a' and a. If a is the root of T, by convention $T_0^{(a)} \bullet_{a'}^{(t_0)} T' = T'$. With this composition, $\mathcal{P}_{\mathcal{T}}$ is an operad in the category of species.

Proof. Note that the tree $\Big(\cdots\Big(\big(T_0^{(a)} \bullet_{\lambda}^{(t_0)} T'\big) \bullet_{v_1}^{(t_1)} T_1^{(a)}\Big)\cdots\Big) \bullet_{v_{n_a}}^{(t_{n_a}^{(a)})} T_{n_a}$, which is shortly denoted by $T \bullet_{\lambda}^{(v)} T'$, is obtained in the following process:

(1) Delete the branches $T_1^{(a)}, \dots, T_{n_a}^{(a)}$ coming from a in T. One obtains a tree T'', and a is a leaf of T''.
(2) Identify $a \in V(T'')$ with the root of T'. One obtains a tree T'''.
(3) Graft $T_1^{(a)}$ on the vertex v_1 of T'''v with an edge of type t_1, …, graft $T_{n_a}^{(a)}$ on the vertex v_{n_a} of T''' with an edge of type t_n. The obtained tree is $T \bullet_\lambda^{(v)} T'$.

This obviously does not depend on the choice of the indexation of $T_1^{(a)}, \dots, T_{n_a}^{(a)}$.

Let $T \in \mathbb{T}_{\mathcal{T}}(A)$, $T' \in \mathbb{T}_{\mathcal{T}}(B)$, $T'' \in \mathbb{T}_{\mathcal{T}}(C)$.

- If $a', a'' \in A$, with $a' \neq a''$, then

$$\begin{aligned}(T \circ_{a'} T') \circ_{a''} T'' &= \sum_{v' \in V(T')^{n_{a'}}, v'' \in V(T'')^{n_{a''}}} \left(T \bullet_{a'}^{(v')} T'\right) \bullet_{a''}^{(v'')} T'' \\ &= \sum_{v' \in V(T')^{n_{a'}}, v'' \in V(T'')^{n_{a''}}} \left(T \bullet_{a''}^{(v'')} T''\right) \bullet_{a'}^{(v')} T' \\ &= (T \circ_{a''} T'') \circ_{a'} T'.\end{aligned}$$

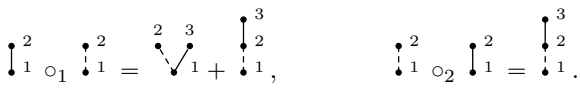
- If $a' \in A$ and $b'' \in B$, then

$$\begin{aligned}(T \circ_{a'} T') \circ_{b''} T'' &= \sum_{v' \in V(T')^{n_{a'}}, v'' \in V(T'')^{n_{b''}}} \left(T \bullet_{a'}^{(v')} T'\right) \bullet_{b''}^{(v'')} T'' \\ &= \sum_{v' \in V(T')^{n_{a'}}, v'' \in V(T'')^{n_{b''}}} T \bullet_{a'}^{(v')} \left(T' \bullet_{b''}^{(v'')} T''\right) \\ &= T \circ_{a'} (T' \circ_{b''} T'').\end{aligned}$$

Moreover, $\bullet^a \bullet_\lambda T = T$ for any tree T, and if $a \in V(T)$, $T \bullet_\lambda \bullet^a T$. So $\mathcal{P}_{\mathbb{T}}$ is indeed an operad in the category of species. □

Consequently, the family $(\mathcal{P}_{\mathcal{T}}(n))_{n \geqslant 0}$ is an operad in the category of vector spaces, which we denote by $\mathcal{P}_{\mathcal{T}}$.

Example 3.15.



Remark 3.16. Another operad on typed trees is introduced in [11]. It is a typed version of the operad of nonassociative, permutative operad of [16], and is different from ours.

In the non-typed case, this theorem is proved in [9]:

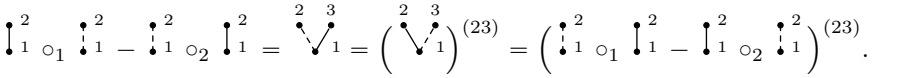
Theorem 3.17. *The operad of $\mathcal{T}$-multiple pre-Lie algebras is isomorphic to $\mathcal{P}_{\mathcal{T}}$, via the isomorphism Φ sending, for any $t \in \mathcal{T}$, $\bullet_t$ to the tree* *, where the edge is of type t.*

Proof. The operad of $\mathcal{T}$-multiple pre-Lie algebras is generated by the binary elements $\bullet_t$, $t \in \mathcal{T}$, with the relations

$$\forall t, t' \in \mathcal{T}, \qquad \bullet_{t'} \circ_2 \bullet_t - \bullet_t \circ_1 \bullet_{t'} = (\bullet_t \circ_2 \bullet_{t'} - \bullet_{t'} \circ_1 \bullet_t)^{(23)},$$

where in the right side we used the action of the symmetric group $\mathfrak{S}_3$ on $\mathcal{T}(3)$, and more specifically the action of the transposition (23). Firstly, if t and t' are

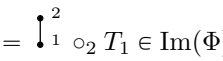
elements of $\mathcal{T}$, symbolized by $\vert$ and $\vdots$, by the preceding example:



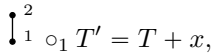
So the morphism ϕ exists. Let us prove that it is surjective: let $T \in \mathbb{T}_{\mathcal{T}}(n)$, we show that it belongs to $\mathrm{Im}(\Phi)$ by induction on n. It is obvious if $n = 1$ or $n = 2$. Let us assume the result at all ranks $< n$. Up to a reindexation we assume that

$$T = B_1(T_1\partial_{t_1} \cdots T_k\partial_{t_k}),$$

where for any $1 \leqslant i < j \leqslant k$, if $x \in V(T_i)$ and $y \in V(T_j)$, then $x < y$. We denote by T_i' the standardization of T_i. By the induction hypothesis on n, $T_i' \in \mathrm{Im}(\Phi)$ for all i. We proceed by induction on k. The type t_k will be represented in red. If $k = 1$, then

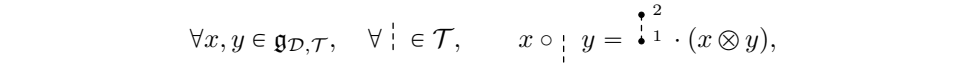


Let us assume the result at rank $k-1$. We put $T' = B_1(T_1\partial_{t_1} \cdots T_{k-1}\partial_{t_{k-1}})$. By the induction hypothesis on n, $T' \in \mathrm{Im}(\Phi)$. Then



where x is a sum of trees with n vertices, such that the fertility of the root is $k-1$. Hence, $x \in \mathrm{Im}(\Phi)$, so $T \in \mathrm{Im}(\Phi)$.

Let $\mathcal{D}$ be a set. The morphism ϕ implies that the free $\mathcal{P}_{\mathbb{T}}$-algebra generated by $\mathcal{D}$, that is to say $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$, inherits a $\mathcal{T}$-multiple pre-Lie algebra structure defined by



where $\cdot$ is the $\mathcal{P}_{\mathcal{T}}$-algebra structure of $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$. For any trees T, T' in $\mathbb{T}_{\mathcal{D},\mathcal{T}}$, by definition of the operadic composition of $\mathcal{P}_{\mathcal{T}}$,

$$T \circ_t T' = \sum_{v \in V(T)} T \bullet_t^{(v)} T',$$

so $\circ_t = \bullet_t$ for any t. As $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, (\bullet_t)_{t\in\mathcal{T}})$ is the free $\mathcal{T}$-multiple pre-Lie algebra generated by $\mathcal{D}$, Φ is an isomorphism. □

Remark 3.18. Let us assume that $\mathcal{T}$ is finite, of cardinality T. Then the components of $\mathcal{P}_{\mathcal{T}}$ are finite-dimensional. As the number of rooted trees which vertices are the elements of $[n]$ is n^{n-1}, for any $n \geqslant 0$ the dimension of $\mathcal{P}_{\mathcal{T}}(n)$ is $T^{n-1}n^{n-1}$, and the formal series of $\mathcal{P}_{\mathcal{T}}$ is

$$f_T(X) = \sum_{n\geqslant 1} \frac{\dim(\mathcal{P}_{\mathcal{T}}(n))}{n!} X^n = \sum_{n\geqslant 1} \frac{(Tn)^{n-1}}{n!} X^n = \frac{f_1(TX)}{T}.$$

3.4. **Koszul dual operad.** If $\mathcal{T}$ is finite, then $\mathcal{P}_{\mathcal{T}}$ is a quadratic operad. Its Koszul dual can be directly computed:

Proposition 3.19. *The Koszul dual operad $\mathcal{P}_{\mathcal{T}}^!$ of $\mathcal{P}_{\mathcal{T}}$ is generated by $\diamond_t$, $t \in \mathcal{T}$, with the relations*

$$\forall t, t' \in \mathcal{T}, \qquad \diamond_{t'} \circ_1 \diamond_t = \diamond_t \circ_2 \diamond_{t'}, \qquad \diamond_{t'} \circ_1 \diamond_t = (\diamond_t \circ_1 \diamond_{t'})^{(23)}.$$

The algebras on $\mathcal{P}_{\mathcal{T}}^!$ are called $\mathcal{T}$-multiple permutative algebras. Such an algebra A is given with the bilinear products $\diamond_t$, $t \in \mathcal{T}$, such that

$$\forall x, y, z \in A, \qquad (x \diamond_t y) \diamond_{t'} z = x \diamond_t (y \diamond_{t'} z), \qquad (x \diamond_t y) \diamond_{t'} z = (x \diamond_{t'} z) \diamond_t y.$$

In particular, for any t, $\diamond_t$ is a permutative product.

Of course, the definition of $\mathcal{T}$-multiple permutative algebras makes sense even if $\mathcal{T}$ is infinite. Permutative algebras are introduced in [8]. If A is a $\mathcal{T}$-multiple permutative algebra, then for any $(\lambda_t)_{t\in\mathcal{T}} \in \mathbb{K}^{(\mathcal{T})}$, $\diamond_a = \sum \lambda_t \diamond_t$ is a permutative product on A.

Proposition 3.20. *Let V be a vector space. Then $V \otimes S\big(V^{\oplus\mathcal{T}}\big)$ is given with a $\mathcal{T}$-multiple permutative algebra structure:*

$$\forall t \in \mathcal{T}, \quad v, v' \in V, \quad w, w' \in S\big(V^{\oplus\mathcal{T}}\big), \qquad (v\otimes w)\diamond_t (v'\otimes w') = v \otimes ww'(v'\partial_t).$$

This $\mathcal{T}$-multiple permutative algebra is denoted by $P_{\mathcal{T}}(V)$. For any $\mathcal{T}$-multiple permutative algebra V and any linear map $\phi\colon V \longrightarrow A$, there exists a unique morphism $\Phi\colon P_{\mathcal{T}}(V) \longrightarrow A$ such that for any $v \in V$, $\Phi(v\otimes 1) = \phi(v)$.

Proof. Let $t, t' \in \mathcal{T}$, $v, v, v'' \in V$, $w, w', w'' \in S\big(V^{\oplus\mathcal{T}}\big)$.

$$\begin{aligned}(v\otimes w\diamond_t v'\otimes w')\diamond_{t'} v''\otimes w'' &= v\otimes w\diamond_t (v'\otimes w'\diamond_{t'} v''\otimes w'')\\ &= (v\otimes w\diamond_t v'\diamond_{t'} v''\otimes w'')\otimes w'\\ &= v\otimes ww'w''(v'\partial_t)(v''\partial_{t'}),\end{aligned}$$

so $P_{\mathcal{T}}(V)$ is $\mathcal{T}$-multiple permutative.

Existence of Φ. Let $t_1,\dots,t_k \in \mathcal{T}$, $v, v_1, \dots, v_k \in V$. We inductively define $\Phi(v\otimes(v_1\partial_{t_1})\cdots(v_k\partial_{t_k}))$ by

$$\Phi(v\otimes 1) = \phi(v),$$

$$\Phi(v\otimes(v_1\partial_{t_1})\cdots(v_k\partial_{t_k})) = \Phi(v\otimes(v_1\partial_{t_1})\cdots(v_{k-1}\partial_{t_{k-1}}))\diamond_{t_k}\phi(v_k) \qquad \text{if} \quad k\geqslant 1.$$

Let us prove that this does not depend on the order chosen on the factors $v_i\partial_{t_i}$ by induction on k. If $k = 0$ or 1, there is nothing to prove. Otherwise, if $i < k$,

$$\begin{aligned}&\Phi(v\otimes(v_1\partial_{t_1})\cdots(v_{i-1}\partial_{t_{i-1}})(v_{i+1}\partial_{t_{i+1}})\cdots(v_k\partial_{t_k}))\diamond_{t_i}\phi(v_i)\\ &= (\Phi(v\otimes(v_1\partial_{t_1})\cdots(v_{i-1}\partial_{t_{i-1}})(v_{i+1}\partial_{t_{i+1}})\cdots(v_{k-1}\partial_{t_{k-1}}))\diamond_{t_k}\phi(v_k))\diamond_{t_i}\phi(v_i)\\ &= (\Phi(v\otimes(v_1\partial_{t_1})\cdots(v_{i-1}\partial_{t_{i-1}})(v_{i+1}\partial_{t_{i+1}})\cdots(v_{k-1}\partial_{t_{k-1}}))\diamond_{t_i}\phi(v_i))\diamond_{t_k}\phi(v_k)\\ &= \Phi(v\otimes(v_1\partial_{t_1})\cdots(v_{k-1}\partial_{t_{k-1}}))\diamond_{t_k}\phi(v_k)\\ &= \Phi(v\otimes(v_1\partial_{t_1})\cdots(v_k\partial_{t_k})).\end{aligned}$$

So Φ is well-defined. Let us prove that Φ is a $\mathcal{T}$-multiple permutative algebra morphism. Let $v, v' \in V$, w, $w' = (v_1\partial_{t_1})\cdots(v_k\partial_{t_k}) \in S\big(V^{\oplus\mathcal{T}}\big)$, and $t \in \mathcal{T}$. Let us prove that $\Phi(v\otimes w\diamond_t v'\otimes w'') = \Phi(v\otimes w)\diamond_t\Phi(v'\otimes w')$ by induction on k. If $k = 0$,

$$\begin{aligned}\Phi(v\otimes w\diamond_t v'\otimes 1) &= \Phi(v\otimes w(v'\partial_t))\\ &= \Phi(v\otimes w)\diamond_t\phi(v')\\ &= \Phi(v\otimes w)\diamond_t\Phi(v'\otimes 1).\end{aligned}$$

Otherwise, we put $w'' = (v_1\partial_{t_1})\cdots(v_{k-1}\partial_{t_{k-1}})$. Then

$$\begin{aligned}\Phi(v\otimes w\diamond_t v'\otimes w') &= \Phi(v\otimes ww''(v'\partial_t)(v_k\partial_{t_k}))\\ &= \Phi(v\otimes ww''(v'\partial_t))\diamond_{t_k}\phi(v_k)\\ &= \Phi(v\otimes w\diamond_t v'\otimes w'')\diamond_{t_k}\phi(v_k)\\ &= (\Phi(v\otimes w)\diamond_t\Phi(v'\otimes w''))\diamond_{t_k}\phi(v_k)\\ &= \Phi(v\otimes w)\diamond_t(\Phi(v'\otimes w'')\diamond_{t_k}\phi(v_k))\\ &= \Phi(v\otimes w')\diamond_t\Phi(v'\otimes w').\end{aligned}$$

So Φ is a $\mathcal{T}$-multiple permutative algebra morphism.

Uniqueness. For any $v, v_1, \dots, v_k \in V$, $t_1, \dots, t_k \in \mathcal{T}$,

$$v \otimes (v_1\partial_{t_1}) \cdots (v_k\partial_{t_k}) = (v \otimes (v_1\partial_{t_1}) \cdots (v_{k-1}\partial_{t_{k-1}})) \diamond_{t_k} v_k.$$

It is then easy to prove that $P_{\mathcal{T}}(V)$ is generated by $V \otimes 1$ as a $\mathcal{T}$-multiple permutative algebra. Consequently, Φ is unique. □

Remark 3.21.

(1) We proved that $P_{\mathcal{T}}(V)$ is freely generated by V, identified with $V \otimes 1$. Consequently, $\mathcal{P}^!_{\mathcal{T}}(n)$ has the same dimension as the multilinear component of $V \otimes S\big(V^{\oplus \mathcal{T}}\big)$ with $V = \mathrm{Vect}(X_1, \dots, X_n)$, that is to say

$$\mathrm{Vect}\big(X_i \otimes (X_1\partial_{t_1}) \cdots (X_{i-1}\partial_{t_{i-1}})(X_{i+1}\partial_{t_{i+1}}) \cdots (X_n\partial_{t_n}),\ 1 \leqslant i \leqslant n, t_j \in \mathcal{T}\big),$$

so

$$\dim(\mathcal{P}^!_{\mathcal{T}}(n)) = nT^{n-1}.$$

The formal series of $\mathcal{P}^!_{\mathcal{T}}$ is

$$f^!_T(X) = \sum_{n \geqslant 1} \frac{\dim(\mathcal{P}^!_{\mathcal{T}}(n))}{n!} X^n = \sum_{n \geqslant 1} \frac{T^{n-1}}{(n-1)!} X^n = X\exp(TX) = \frac{f^!_1(TX)}{T}.$$

(2) It is possible to prove that $\mathcal{P}^!_{\mathcal{T}}$ is a Koszul operad (and, hence, $\mathcal{P}_{\mathcal{T}}$ too) using the rewriting method of [17].

4. Structure of the pre-Lie products

4.1. **A nonassociative permutative coproduct.** For all $t \in \mathcal{T}$, we define a coproduct $\rho_t \colon \mathfrak{g}_{\mathcal{D},\mathcal{T}} \longrightarrow \mathfrak{g}^{\otimes 2}_{\mathcal{D},\mathcal{T}}$ by

$$\forall T = B_d\Big(\prod_{i \in [k]} T_i\partial_{t_i}\Big) \in \mathbb{T}_{\mathcal{D},\mathcal{T}}, \qquad \rho_t(T) = \sum_{j \in [k]} \delta_{t,t_j} B_d\Big(\prod_{i \in [k],\ i \neq j} T_i\partial_{t_i}\Big) \otimes T_j\partial_t.$$

Proposition 4.1.

1. *For all $t, t' \in \mathcal{T}$, $(\rho_t \otimes \mathrm{Id}) \circ \rho_{t'} = ((\rho_{t'} \otimes \mathrm{Id}) \circ \rho_t)^{(23)}$, where in the right side we use the classical action of the symmetric group $\mathfrak{S}_3$ on $\mathfrak{g}^{\otimes 3}_{\mathcal{D},\mathcal{T}}$ by permutations of the tensors, and more specifically the action of the transposition (23). With Sweedler's notation for ρ_t, this can be written, for any $x \in \mathfrak{g}_{\mathcal{D},\mathcal{T}}$, as*

$$\sum\sum \big(x^{(1)_{t'}}\big)^{(1)_t} \otimes \big(x^{(1)_{t'}}\big)^{(2)_t} \otimes x^{(2)_{t'}} = \sum\sum \big(x^{(1)_t}\big)^{(1)_{t'}} \otimes x^{(2)_t} \otimes \big(x^{(1)_t}\big)^{(2)_{t'}}.$$

2. *For any $x, y \in \mathfrak{g}_{\mathcal{D},\mathcal{T}}$, $t, t' \in \mathcal{T}$, with Sweedler's notations $\rho_t(x) = \sum x^{(1)_t} \otimes x^{(2)_t}$,*

$$\rho_t(x \bullet_{t'} y) = \delta_{t,t'} x \otimes y + \sum x^{(1)_t} \bullet_{t'} y \otimes x^{(2)_t} + \sum x^{(1)_t} \otimes x^{(2)_t} \bullet_{t'} y.$$

3. *For any $\mu = (\mu_t)_{t \in \mathcal{T}} \in \mathbb{K}^{\mathcal{T}}$, we put*

$$\rho_\mu = \sum_{t \in \mathcal{T}} \mu_t \rho_t \colon\ \mathfrak{g}_{\mathcal{D},\mathcal{T}} \longrightarrow \mathfrak{g}^{\otimes 2}_{\mathcal{D},\mathcal{T}}.$$

This makes sense, as any tree in $\mathbb{T}_{\mathcal{D},\mathcal{T}}$ does not vanish only under a finite number of ρ_t. Then ρ_μ is a nonassociative permutative (NAP) coproduct; for any $x, y \in \mathfrak{g}_{\mathcal{D},\mathcal{T}}$, by the second point, using Sweedler's notation for ρ_μ,

$$\rho_\mu(x \bullet_\lambda y) = \Big(\sum_{t \in \mathcal{T}} \lambda_t\mu_t\Big) x \otimes y + \sum x^{(1)_\mu} \bullet_\lambda y \otimes x^{(2)_\mu} + \sum x^{(1)_\mu} \otimes x^{(2)_\mu} \bullet_\lambda y.$$

In particular, if $\sum_{t \in \mathcal{T}} \lambda_t\mu_t = 1$, $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda, \rho_\mu)$ is a NAP pre-Lie bialgebra in the sense of [16].

Proof. 1. For any tree T,

$$(\rho_t \otimes \mathrm{Id}) \circ \rho_{t'}(T) = \sum_{p,q\in[k],p\neq q} \delta_{t_p,t}\delta_{t_q,t'} B_d\Big(\prod_{i\in[k],i\neq p,q} T_i\partial_{t_i}\Big) \otimes T_p\partial_t \otimes T_q\partial_{t'},$$

which implies the result.

2. For any tree T, T',

$$\begin{aligned}
\rho_t(T \bullet_{t'} T') &= \rho_t\Big(B_d\Big(\prod_{i\in[k]} T_i\partial_{t_i}T'\partial_{t'}\Big) + \sum_{i\in[k]} B_d\Big(\prod_{j\in[k],j\neq i} T_j\partial_{t_j}(T_i \bullet_{t'} T')\partial_{t_i}\Big)\Big) \\
&= \delta_{t,t'}B_d\Big(\prod_{i\in[k]} T_i\partial_{t_i}\Big)\otimes T' + \sum_{i\in[k]} \delta_{t_i,t}B_d\Big(\prod_{j\in[k],j\neq i} T_j\partial_{t_j}T'\partial_{t'}\Big)\otimes T_i\partial_t \\
&\quad + \sum_{i\in[k]} \delta_{t_i,t}B_d\Big(\prod_{j\in[k],j\neq i} T_j\partial_{t_j}\Big)\otimes (T_i \bullet_{t'} T')\partial_t \\
&\quad + \sum_{i\neq j\in[k]} \delta_{t_i,t}B_d\Big(\prod_{p\in[k],p\neq i,j} T_p\partial_{t_p}(T_j \bullet_{t'} T')\partial_{t_j}\Big)\otimes T_i\partial_t \\
&= \delta_{t,t'}T\otimes T' + \sum_{i\in[k]} \delta_{t_i,t}B_d\Big(\prod_{j\in[k],j\neq i} T_j\partial_{t_j}\Big) \bullet_{t'} T'\otimes T_i\partial_t \\
&\quad + \sum_{i\in[k]} \delta_{t_i,t}B_d\Big(\prod_{j\in[k],j\neq i} T_j\partial_{t_j}\Big)\otimes T_i \bullet_{t'} T'\partial_t \\
&= \delta_{t,t'}T\otimes T' + T^{(1)_t} \bullet_{t'} T'\otimes T^{(2)_t} + T^{(1)_t}\otimes T^{(2)_t} \bullet_{t'} T'.
\end{aligned}$$

3. Obtained by summation. □

Corollary 4.2. *If $\lambda \in \mathbb{K}^{(\mathcal{T})}$ is nonzero, let us choose $t_0 \in \mathcal{T}$ such that $\lambda_{t_0} \neq 0$. The pre-Lie algebra $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$ is freely generated by the set $\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$ of $\mathcal{T}$-typed $\mathcal{D}$-decorated trees T such that there is no edge outgoing the root of T of type t_0.*

Proof. For any tree T, we denote by α_T the number of edges outgoing the root of T of type T_0. Our aim is to prove that $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$ is freely generated by the trees T such that $\alpha_T = 0$. We define a family of scalars $(\mu_t)_{t\in\mathcal{T}}$ by

$$\forall t \in \mathcal{T}, \qquad \mu_t = \begin{cases} 0 & \text{if } t \neq t_0, \\ \dfrac{1}{\lambda_{t_0}} & \text{if } t = t_0. \end{cases}$$

Note that $\rho_\mu = \frac{1}{\lambda_{t_0}}\rho_{t_0}$. By Proposition 4.1, $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda, \rho_\mu)$ is a NAP pre-Lie bialgebra, so by Livernet's rigidity theorem [16], it is freely generated by $\mathrm{Ker}(\rho_\mu) = \mathrm{Ker}(\rho_{t_0})$. Obviously, if $\alpha_T = 0$, $T \in \mathrm{Ker}(\rho_{t_0})$. Let us consider $x = \sum_{T\in\mathbb{T}_{\mathcal{D},\mathcal{T}}} x_T T \in \mathrm{Ker}(\rho_{t_0})$. We consider the map

$$\Upsilon\colon \begin{cases} \mathfrak{g}_{\mathcal{D},\mathcal{T}} \otimes \mathfrak{g}_{\mathcal{D},\mathcal{T}} \longrightarrow \mathfrak{g}_{\mathcal{D},\mathcal{T}}, \\ T\otimes T' \longmapsto T \bullet_{t_0}^{\mathrm{root}(T)} T'. \end{cases}$$

By definition of ρ_{t_0}, for any tree T, $\Upsilon \circ \rho_{t_0}(T) = \alpha_T T$. Consequently,

$$0 = \Upsilon \circ \rho_{t_0}(x) = \sum_{T\in\mathbb{T}_{\mathcal{D},\mathcal{T}}} x_T\alpha_T T.$$

So if $\alpha_T \neq 0$, $x_T = 0$, and x is a linear span of trees such that $\alpha_T = 0$: the set of trees T such that $\alpha_T = 0$ is a basis of $\mathrm{Ker}(\rho_{t_0})$. □

If $|\mathcal{D}| = D$ and $|\mathcal{T}| = T$, the number of elements of $\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$ of degree n is denoted by $t'_{D,T}(n)$; it does not depend on t_0. By direct computations,

$$t'_{D,T}(1) = D,$$
$$t'_{D,T}(2) = D^2(T-1),$$
$$t'_{D,T}(3) = \frac{D^2(T-1)(3DT - D + 1)}{2},$$
$$t'_{D,T}(4) = \frac{D^2(T-1)\big(16D^2T^2 - 8D^2T + D^2 + 6DT - 3D + 2\big)}{6}.$$

In the particular case $D = 1$, $T = 2$, we recover sequence A005750 of the OEIS.

4.2. **Pre-Lie algebra morphisms.**

Notation 4.3. Let $\mathcal{T}$ and $\mathcal{T}'$ be two sets of types. We denote by $\mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$ the space of matrices $M = (m_{t,t'})_{(t,t')\in\mathcal{T}\times\mathcal{T}'}$, such that for any $t' \in \mathcal{T}'$, $(m_{t,t'})_{t\in\mathcal{T}} \in \mathbb{K}^{(\mathcal{T})}$. If $\mathcal{T} = \mathcal{T}'$, we shall simply write $\mathcal{M}_{\mathcal{T}}(\mathbb{K})$. If $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$ and $M' \in M_{\mathcal{T}',\mathcal{T}''}(\mathbb{K})$, then

$$MM' = \left(\sum_{t'\in\mathcal{T}'} m_{t,t'}m'_{t',t''}\right)_{(t,t'')\in\mathcal{T}\times\mathcal{T}''} \in \mathcal{M}_{\mathcal{T},\mathcal{T}''}(\mathbb{K}).$$

If $\lambda \in \mathbb{K}^{(\mathcal{T}')}$ and $\mu \in \mathbb{K}^{\mathcal{T}}$, then

$$M\lambda = \left(\sum_{t'\in\mathcal{T}'} m_{t,t'}\lambda_{t'}\right)_{t\in\mathcal{T}} \in \mathbb{K}^{(\mathcal{T})}, \qquad M^\top\mu = \left(\sum_{t\in\mathcal{T}} m_{t,t'}\mu_t\right)_{t'\in\mathcal{T}'} \in \mathbb{K}^{\mathcal{T}'}.$$

In particular, $\mathcal{M}_{\mathcal{T}}(\mathbb{K})$ is an algebra, acting on $\mathbb{K}^{(\mathcal{T})}$ on the left and on $\mathbb{K}^{\mathcal{T}}$ on the right.

Definition 4.4. Let $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$. We define a map $\Phi_M \colon \mathcal{H}_{\mathcal{D},\mathcal{T}'} \longrightarrow \mathcal{H}_{\mathcal{D},\mathcal{T}}$, sending $F \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$ to the forest obtained by replacing $\mathrm{type}(e)$ by $\sum_{t\in\mathcal{T}} m_{t,\mathrm{type}(e)}t$ for any $e \in E(F)$, F being considered as linear in any of its edges. The restriction of Φ_M to $\mathfrak{g}_{\mathcal{D},\mathcal{T}'}$ is denoted by $\phi_M \colon \mathfrak{g}_{\mathcal{D},\mathcal{T}'} \longrightarrow \mathfrak{g}_{\mathcal{D},\mathcal{T}}$.

Example 4.5. If $\mathcal{T}$ contains two elements, the first one represented by $\mid$ and the second one by $\vdots$, if $M = \left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)$, for any $x, y, z \in \mathcal{D}$,

$$\phi_M\Big(\begin{array}{c}\bullet^{\,y}\\ \mid\\ \bullet_{\,x}\end{array}\Big) = \alpha\begin{array}{c}\bullet^{\,y}\\ \mid\\ \bullet_{\,x}\end{array} + \gamma\begin{array}{c}\bullet^{\,y}\\ \vdots\\ \bullet_{\,x}\end{array},$$

$$\phi_M\Big(\begin{array}{c}\bullet^{\,y}\\ \vdots\\ \bullet_{\,x}\end{array}\Big) = \beta\begin{array}{c}\bullet^{\,y}\\ \mid\\ \bullet_{\,x}\end{array} + \delta\begin{array}{c}\bullet^{\,y}\\ \vdots\\ \bullet_{\,x}\end{array},$$

$$\phi_M\Big(\begin{array}{ccc}y & & z\\ \bullet & & \bullet\\ & \diagdown\!\diagup & \\ & \bullet_{\,x} & \end{array}\Big) = \alpha\beta\begin{array}{ccc}y & & z\\ \bullet & & \bullet\\ & \diagdown\!\diagup & \\ & \bullet_{\,x} & \end{array} + \alpha\delta\begin{array}{ccc}y & & z\\ \bullet & & \bullet\\ & \diagdown\!\diagup & \\ & \bullet_{\,x} & \end{array} + \beta\gamma\begin{array}{ccc}y & & z\\ \bullet & & \bullet\\ & \diagdown\!\diagup & \\ & \bullet_{\,x} & \end{array} + \gamma\delta\begin{array}{ccc}y & & z\\ \bullet & & \bullet\\ & \diagdown\!\diagup & \\ & \bullet_{\,x} & \end{array}.$$

Remark 4.6. For any $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$, $M' \in \mathcal{M}_{\mathcal{T}',\mathcal{T}''}(\mathbb{K})$, $\Phi_M \circ \Phi_{M'} = \Phi_{MM'}$.

Proposition 4.7. *Let $\lambda \in \mathbb{K}^{(\mathcal{T})}$, $\mu \in \mathbb{K}^{\mathcal{T}}$ and $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$. Then ϕ_M is a pre-Lie morphism from $(\mathfrak{g}_{\mathcal{D},\mathcal{T}'}, \bullet_\lambda)$ to $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{M\lambda})$ and a NAP coalgebra morphism from $(\mathfrak{g}_{\mathcal{D},\mathcal{T}'}, \rho_{M^\top\mu})$ to $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \rho_\mu)$.*

Proof. Let $T, T' \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$. For any $t \in \mathcal{T}$, for any $v \in V(T)$,

$$\phi_M\big(T \bullet_t^{(v)} T'\big) = \sum_{t'\in\mathcal{T}} m_{t',t}\phi_M(T) \bullet_{t'} \phi_M(T'),$$

so

$$\phi_M(T \bullet_\lambda T') = \sum_{t,t'\in\mathcal{T}} m_{t',t}\lambda_t\phi_M(T) \bullet_{t'} \phi_M(T') = \phi_M(T) \bullet_{M\lambda} \phi_M(T').$$

We proved that ϕ_M is a pre-Lie algebra morphism from $(\mathfrak{g}_{\mathcal{D},\mathcal{T}'}, \bullet_\lambda)$ to $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{M\lambda})$.

For any $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$,

$$\rho_t \circ \phi_M(T) = \sum_{t' \in \mathcal{T}} m_{t,t'}(\phi_M \otimes \phi_M) \circ \rho_{t'}(T),$$

so

$$\rho_\mu \circ \phi_M(T) = \sum_{t,t' \in \mathcal{T}} m_{t,t'}\mu_t(\phi_M \otimes \phi_M) \circ \rho_{t'}(T) = (\phi_M \otimes \phi_M) \circ \rho_{M^\top \mu}(T).$$

So $\phi_M \colon (\mathfrak{g}_{\mathcal{D},\mathcal{T}'}, \rho_{M^\top \mu}) \longrightarrow (\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \rho_\mu)$ is a NAP coalgebra morphism. □

Corollary 4.8. *For any $\lambda \in \mathbb{K}^{(\mathcal{T})}$ and $\mu \in \mathbb{K}^{\mathcal{T}}$, such that $\sum_{t \in \mathcal{T}} \lambda_t \mu_t = 1$, for any $t_0 \in \mathcal{T}$, the NAP pre-Lie bialgebras $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda, \rho_\mu)$ and $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{t_0}, \rho_{t_0})$ are isomorphic.*

Proof. Let us denote by $\lambda^{(0)}$ the element of $\mathbb{K}^{(\mathcal{T})}$ defined by

$$\lambda^{(0)}_t = \delta_{t,t_0}.$$

Note that for any $M \in \mathcal{M}_{\mathcal{T}}(\mathbb{K})$, invertible, $\phi_M \colon (\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{\lambda^{(0)}}, \rho_{M^\top \mu}) \longrightarrow (\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{M\lambda^{(0)}}, \rho_\mu)$ is an isomorphism. In particular, for a well-chosen M, $M\lambda^{(0)} = \lambda$; we can assume that $\lambda = \lambda^{(0)}$ without loss of generality. Then, by hypothesis, $\mu_{t_0} = 1$. We define a matrix $M \in \mathcal{M}_{\mathcal{T}}(\mathbb{K})$ in the following way:

$$m_{t,t'} = \begin{cases} \delta_{t,t_0} & \text{if } t' = t_0, \\ \delta_{t,t'} - \mu_{t'}\delta_{t,t_0} & \text{otherwise.} \end{cases}$$

Then M is invertible. Moreover, $M\lambda^{(0)} = \lambda^{(0)}$ and $M^\top \mu = \lambda^{(0)}$. So ϕ_M is an isomorphism from $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_{\lambda^{(0)}}, \rho_{\lambda^{(0)}})$ to $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda, \rho_\mu)$. □

Proposition 4.9. *Let $\lambda \in \mathbb{K}^{(\mathcal{T})}$, and $t_0 \in \mathcal{T}$. We define a pre-Lie algebra morphism $\psi_{t_0} \colon (\mathfrak{g}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}, \bullet) \longrightarrow (\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$, sending $\bullet^T$ to T for any $T \in \mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$. Then ψ_{t_0} is a pre-Lie algebra isomorphism if, and only if, $\lambda_{t_0} \neq 0$.*

Proof. If $\lambda_{t_0} \neq 0$, then by corollary 4.2, $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$ is freely generated by $\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$, so ψ_{t_0} is an isomorphism. If $\lambda_{t_0} = 0$, then it is not difficult to show that any tree T with two vertices, with its unique edge of type t_0, does not belong to $\mathrm{Im}(\psi_{t_0})$. □

5. Hopf algebraic structures

We here fix $\lambda \in \mathbb{K}^{(\mathcal{T})}$.

5.1. **Enveloping algebra of $\mathfrak{g}_{\mathcal{D},\mathcal{T}}$.** Using again the Guin–Oudom construction, we obtain the enveloping algebra of $(\mathfrak{g}_{\mathcal{D},\mathcal{T}}, \bullet_\lambda)$. We first identify the symmetric coalgebra $S(\mathfrak{g}_{\mathcal{D},\mathcal{T}})$ with the vector space generated by $\mathbb{F}_{\mathcal{D},\mathcal{T}}$, which we denote by $\mathcal{H}_{\mathcal{D},\mathcal{T}}$. Its product m is given by disjoint union of forests, its coproduct by

$$\forall T_1, \dots, T_k \in \mathbb{T}_{\mathcal{D},\mathcal{T}}, \qquad \Delta(T_1 \cdots T_n) = \sum_{I \subseteq [n]} \prod_{i \in I} T_i \otimes \prod_{i \notin I} T_i.$$

We denote by $\bullet_\lambda$ the Guin–Oudom extension of $\bullet_\lambda$ to $\mathcal{H}_{\mathcal{D},\mathcal{T}}$ and $\star_\lambda$ the associated associative product.

Theorem 5.1. *For any $F \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$, $T_1, \dots, T_n \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$,*

$$F \bullet_\lambda T_1 \cdots T_n = \sum_{\substack{v_1,\dots,v_n \in V(F), \\ t_1,\dots,t_n \in \mathcal{T}}} \left(\prod_{i \in [n]} \lambda_{t_i} \right) \left(\cdots \left(F \bullet_{t_1}^{(v_1)} T_1 \right) \cdots \right) \bullet_{t_n}^{(v_n)} T_n,$$

$$F \star_\lambda T_1 \cdots T_n = \sum_{I \subseteq [n]} \left(F \bullet_\lambda \prod_{i \in I} T_i \right) \prod_{i \notin I} T_i.$$

The Hopf algebra $(\mathcal{H}_{\mathcal{D},\mathcal{T}}, \star_\lambda, \Delta)$ is denoted by $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{GL}_\lambda}$. Moreover, for any $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$, for any $\lambda \in \mathbb{K}^{(\mathcal{T}')}$, Φ_M is a Hopf algebra morphism from $\mathcal{H}_{\mathcal{D},\mathcal{T}'}^{\mathrm{GL}_\lambda}$ to $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{GL}_{M\lambda}}$. The extension of ψ_{t_0} as a Hopf algebra morphism from $\mathcal{H}_{\mathbb{T}_{\mathcal{D},\mathcal{T}}^{(t_0)}}^{\mathrm{GL}}$ to $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{GL}_\lambda}$ is denoted by Ψ_{t_0}; it is an isomorphism if, and only if, $\lambda_{t_0} \neq 0$.

In particular, if $\mathcal{T} = \{t\}$ and $\lambda_t = 1$, we recover the Grossman–Larson Hopf algebra [14].

5.2. **Dual construction.**

Proposition 5.2. *Let $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$.*

1. *A cut c of T is a nonempty subset of $E(T)$; it is said to be admissible if any path in the tree from the root to a leaf meets at most one edge in c. The set of admissible cuts of T is denoted by $\mathrm{Adm}(T)$.*
2. *If c is admissible, one of the connected components of $T\backslash c$ contains the root of c: we denote it by $R^c(T)$. The product of the other connected components of $T\backslash c$ is denoted by $P^c(T)$.*

Let $\lambda \in \mathbb{K}^{\mathcal{T}}$. We define a multiplicative coproduct $\Delta^{\mathrm{CK}_\lambda}$ on the algebra $(\mathcal{H}_{\mathcal{D},\mathcal{T}}, m)$ by

$$\forall T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}, \qquad \Delta^{\mathrm{CK}_\lambda}(T) = T \otimes 1 + 1 \otimes T + \sum_{c \in \mathrm{Adm}(T)} \left(\prod_{e \in c} \lambda_{\mathrm{type}(e)} \right) R^c(T) \otimes P^c(T).$$

Then $\left(\mathcal{H}_{\mathcal{D},\mathcal{T}}, m, \Delta^{\mathrm{CK}_\lambda}\right)$ is a Hopf algebra, which we denote by $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\lambda}$.

Proof. We first assume that $\lambda \in \mathbb{K}^{(\mathcal{T})}$. Let us define a nondegenerate pairing $\langle -, - \rangle$ on $\mathcal{H}_{\mathcal{D},\mathcal{T}}$ by

$$\forall F, F' \in \mathbb{F}_{\mathcal{D},\mathcal{T}}, \qquad \langle F, F' \rangle = \delta_{F,F'} s_F,$$

where s_F is the number of symmetries of F. Let us consider three forests F, F', F''. We put

$$F = \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} T^{\lambda_t}, \qquad F' = \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} T^{a'_T}, \qquad F'' = \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} T^{a''_T}.$$

Then

$$\begin{aligned}
\langle \Delta(F), F' \otimes F'' \rangle &= \sum_{a=b+c} \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} \frac{\lambda_t!}{\mu_t! c_T!} \left\langle \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} T^{\mu_t}, F' \right\rangle \left\langle \prod_{T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}} T^{c_T}, F'' \right\rangle \\
&= \sum_{a=b+c} \delta_{b,a'} \delta_{c,a''} \frac{\lambda_t!}{a'_T! a''_T!} s_{F'} s_{F''} = \delta_{a,a'+a''} \frac{\lambda_t!}{a'_T! a''_T!} a'_T! a''_T! s_T^{a'_T + a''_T} \\
&= \delta_{a,a'+a''} \lambda_t! s_T^{\lambda_t} = \delta_{F,F'F''} s_F \\
&= \langle F, F'F'' \rangle.
\end{aligned}$$

Therefore,

$$\forall x, y, z \in \mathcal{H}_{\mathcal{D},\mathcal{T}}, \qquad \langle \Delta(x), y \otimes z \rangle = \langle x, yz \rangle.$$

Let F, G be two forests and T be a tree. Observe that if F is a forest with at least two trees, then $F \star_\lambda G$ does not contain any tree, so $\langle F \star_\lambda G, T \rangle = 0$. If $F = 1$, then $\langle F \star_\lambda G, T \rangle \neq 0$ if, and only if, $G = T$; moreover, $\langle 1 \star_\lambda T, T \rangle = 1$. If F is a tree, then

$$\langle F \star_\lambda G, T \rangle = \langle F \bullet_\lambda G, T \rangle.$$

Moreover, if $F = B_d(F')$ and $G = T_1 \cdots T_k$,

$$F \bullet_\lambda G = \sum_{I \subseteq [k]} \sum_{(t_i) \in \mathcal{T}^k} \Big(\prod_{i \in [k]} \lambda_{t_i} \Big) B_d\Big(\prod_{i \in I} T_i \partial_{t_i} F' \bullet \prod_{i \notin I} T_i \partial_{t_i} \Big),$$

where $\bullet$ is the pre-Lie product on $\mathfrak{g}^{\mathcal{T}}_{\mathcal{D},\mathcal{T}}$ induced by the $\mathcal{T}$-multiplie pre-Lie structure. Consequently, we can inductively define a coproduct $\Delta^{\mathrm{CK}_\lambda} \colon \mathcal{H}_{\mathcal{D},\mathcal{T}} \longrightarrow \mathcal{H}_{\mathcal{D},\mathcal{T}} \otimes \mathcal{H}_{\mathcal{D},\mathcal{T}}$, multiplicative for the product m, such that, if we denote for any tree T, $\overline{\Delta}_{\mathrm{CK}}(T) = \Delta(T) - 1 \otimes T$, for any tree $T = B_d(T_1 \partial_{t_1} \cdots T_k \partial_{t_k})$,

$$\overline{\Delta}^{\mathrm{CK}}_\lambda(T) = (B_d \otimes \mathrm{Id}) \Big(\prod_{i \in [k]} \Big(\overline{\Delta}^{\mathrm{CK}}_\lambda(T_i) \partial_{t_i} \otimes 1 + \lambda_{t_i} 1 \otimes T_i \Big) \Big).$$

Then, for any $x, y, z \in \mathcal{H}_{\mathcal{D},\mathcal{T}}$,

$$\big\langle x \star_\lambda y, z \big\rangle = \big\langle x \otimes y, \Delta^{\mathrm{CK}_\lambda}(z) \big\rangle.$$

A quite easy induction on the number of vertices of trees proves that this coproduct is indeed the one we define in the statement of the proposition. As $\langle -, - \rangle$ is nondegenerate, $\big(\mathcal{H}_{\mathcal{D},\mathcal{T}}, m, \Delta^{\mathrm{CK}_\lambda}\big)$ is a Hopf algebra, dual to $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}}$.

In the general case, for any $x \in \mathcal{H}_{\mathcal{D},\mathcal{T}}$, there exists a finite subset $\mathcal{T}'$ of $\mathcal{T}$ such that $x \in \mathcal{H}_{\mathcal{D},\mathcal{T}'}$. Putting $\lambda' = \lambda_{|\mathcal{T}'}$, $\lambda' \in \mathbb{K}^{\mathcal{T}'} = \mathbb{K}^{(\mathcal{T}')}$, so

$$\begin{aligned} \big(\Delta^{\mathrm{CK}_\lambda} \otimes \mathrm{Id}\big) \circ \Delta^{\mathrm{CK}_\lambda}(x) &= \big(\Delta^{\mathrm{CK}_{\lambda'}} \otimes \mathrm{Id}\big) \circ \Delta^{\mathrm{CK}_{\lambda'}}(x) \\ &= \big(\mathrm{Id} \otimes \Delta^{\mathrm{CK}_{\lambda'}}\big) \circ \Delta^{\mathrm{CK}_{\lambda'}}(x) \\ &= \big(\mathrm{Id} \otimes \Delta^{\mathrm{CK}_\lambda}\big) \circ \Delta^{\mathrm{CK}_\lambda}(x). \end{aligned}$$

Hence, Δ_λ is coassociative, and $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ is a Hopf algebra. □

Example 5.3. Let us fix a subset $\mathcal{T}'$ of $\mathcal{T}$ and choose $(\lambda_t)_{t \in \mathcal{T}}$ such that

$$\lambda_t = \begin{cases} 1 & \text{if } t \in \mathcal{T}', \\ 0 & \text{otherwise.} \end{cases}$$

For any tree $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$, let us denote by $\mathrm{Adm}_{\mathcal{T}'}(T)$ the set of admissible cuts c of T such that the type of any edge in c belongs to $\mathcal{T}'$. Then

$$\Delta^{\mathrm{CK}_\lambda}(T) = T \otimes 1 + 1 \otimes T + \sum_{c \in \mathrm{Adm}_{\mathcal{T}'}(T)} R^c(T) \otimes P^c(T).$$

Remark 5.4.

(1) If $\mathcal{T} = \{t\}$ and $\lambda_t = 1$, we recover the usual Connes–Kreimer Hopf algebra of $\mathcal{D}$-decorated rooted trees, which we denote by $\mathcal{H}^{\mathrm{CK}}_{\mathcal{D}}$, and its duality with the Grossman–Larson Hopf algebra [10, 15, 23].

(2) If $\mathcal{T}$ and $\mathcal{D}$ are finite, for any $\lambda \in \mathbb{K}^{\mathcal{T}}$, both $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}}$ are graded Hopf algebras (by the number of vertices), and their homogeneous components are finite-dimensional. Via the pairing $\langle -, - \rangle$, each one is the graded dual of the other.

5.3. **Hochschild cohomology of coalgebras.** For the sake of simplicity, we assume that the set of types $\mathcal{T}$ is finite and we put $\mathcal{T} = \{t_1, \ldots, t_N\}$.

Let (C, Δ) be a coalgebra and let (M, δ_L, δ_R) be a C-bicomodule. We define a complex, dual to the Hochschild complex for algebras, in the following way:

(1) For any $n \geqslant 0$, $H_n = \mathcal{L}\big(M, C^{\otimes n}\big)$.

(2) For any $L \in H_n$,

$$b_n(L) = (\mathrm{Id} \otimes L) \circ \delta_L + \sum_{i=1}^{n} (-1)^i \big(\mathrm{Id}^{\otimes(i-1)} \otimes \Delta \otimes \mathrm{Id}^{\otimes(n-i)}\big) \circ L + (-1)^{n+1} (L \otimes \mathrm{Id}) \circ \delta_R.$$

In particular, one-cocycles are maps $L\colon M \longrightarrow C$ such that

$$\Delta \circ L = (\mathrm{Id} \otimes L) \circ \delta_L + (L \otimes \mathrm{Id}) \circ \delta_R.$$

We shall consider in particular the bicomodule (M, δ_L, δ_R) such that

$$\forall x \in C, \qquad \begin{cases} \delta_L(x) = 1 \otimes x, \\ \delta_R(x) = \Delta(x). \end{cases}$$

If C is a bialgebra, then $M^{\otimes N}$ is also a bicomodule:

$$\forall x_t \in C, \qquad \begin{cases} \delta_L\Big(\bigotimes\limits_{1 \leqslant i \leqslant N} x_i\Big) = 1 \otimes \bigotimes\limits_{1 \leqslant i \leqslant N} x_i, \\ \delta_R\Big(\bigotimes\limits_{1 \leqslant i \leqslant N} x_i\Big) = \bigotimes\limits_{1 \leqslant i \leqslant N} x_i^{(1)} \otimes \prod\limits_{1 \leqslant i \leqslant N} x_i^{(2)}. \end{cases}$$

We denote by $\underline{1} = (1)_{t \in \mathcal{T}} \in \mathbb{K}^{\mathcal{T}}$, and we take $C = \mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_{\underline{1}}}$. One can identify $S\big(\mathrm{Vect}(\mathbb{T}_{\mathcal{D},\mathcal{T}})^{\oplus \mathcal{T}}\big)$ and $C^{\otimes N}$, $x\delta_{T_i}$ being identified with $1^{\otimes(i-1)} \otimes x \otimes 1^{\otimes(n-i)}$ for any $x \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$ and $1 \leqslant i \leqslant N$. Then for any d, $B_d\colon C^{\otimes N} \longrightarrow C$ is a 1-cocycle. Moreover, there is a universal property, proved in the same way as for the Connes–Kreimer's one [10]:

Theorem 5.5. *Let B be a commutative bialgebra and, for any $d \in \mathcal{D}$, let $L_d\colon C^{\otimes N} \longrightarrow C$ be a 1-cocycle:*

$$\forall d \in \mathcal{D}, \quad \forall x_t \in B, \qquad \Delta \circ L_d\Big(\bigotimes_{1 \leqslant i \leqslant N} x_i\Big) = 1 \otimes \bigotimes_{1 \leqslant i \leqslant N} x_i + L_d\Big(\bigotimes_{1 \leqslant i \leqslant N} x_i^{(1)}\Big) \otimes \prod_{1 \leqslant i \leqslant N} x_i^{(2)}.$$

There exists a unique bialgebra morphism $\phi\colon \mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_{\underline{1}}} \longrightarrow C$ such that for any $d \in \mathcal{D}$, $\phi \circ L_d = B_d \circ \phi^{\otimes N}$.

5.4. **Hopf algebra morphisms.** Our aim is, firstly, to construct Hopf algebras morphisms between $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\lambda}$ and $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\mu}$; secondly, to construct Hopf algebra isomorphisms between $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\lambda}$ and $\mathcal{H}_{\mathcal{D}'}^{\mathrm{CK}}$ for a well-chosen $\mathcal{D}'$.

Proposition 5.6. *Let $M \in \mathcal{M}_{\mathcal{T},\mathcal{T}'}(\mathbb{K})$, $\lambda \in \mathbb{K}^{\mathcal{T}}$. Then $\Phi_M\colon \mathcal{H}_{\mathcal{D},\mathcal{T}'}^{\mathrm{CK}_{M^\top \lambda}} \longrightarrow \mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\lambda}$ is a Hopf algebra morphism.*

Proof. Φ_M is obviously an algebra morphism. Let $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$.

$$\begin{aligned} \Delta_\lambda \circ \Phi_M(T) &= \Phi_M(T) \otimes 1 + 1 \otimes \Phi_M(T) \\ &+ \sum_{c \in \mathrm{Adm}(T)} \prod_{e \in c} \Big(\sum_{t \in \mathcal{T}} m_{t,\mathrm{type}(e)} \lambda_t \Big) \Phi_M(R^c(T)) \otimes \Phi_M(P^c(T)) \\ &= \Phi_M(T) \otimes 1 + 1 \otimes \Phi_M(T) \\ &+ \sum_{c \in \mathrm{Adm}(T)} \prod_{e \in c} \big(M^\top \lambda\big)_{\mathrm{type}(e)} \Phi_M(R^c(T)) \otimes \Phi_M(P^c(T)) \\ &= (\Phi_M \otimes \Phi_M) \circ \Delta_{M^\top a}(T). \end{aligned}$$

So Φ_M is a coalgebra morphism from $\mathcal{H}_{\mathcal{D},\mathcal{T}'}^{\mathrm{CK}_{M^\top \lambda}}$ to $\mathcal{H}_{\mathcal{D},\mathcal{T}}^{\mathrm{CK}_\lambda}$. □

Corollary 5.7. *Let $\lambda, \mu \in \mathbb{K}^{\mathcal{T}}$, both nonzero. Then $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{CK}_\mu}_{\mathcal{D},\mathcal{T}}$ are isomorphic Hopf algebras.*

Proof. There exists $M \in \mathcal{M}_{\mathcal{T}}(\mathbb{K})$, invertible, such that $M^\top \lambda = \mu$. Then Φ_M is an isomorphism between $\mathcal{H}^{\mathrm{CK}_\mu}_{\mathcal{D},\mathcal{T}}$ and $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$. □

Definition 5.8. Let us fix $t_0 \in \mathcal{T}$. For any $F \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$, we shall say that $\{T_1, \dots, T_k\} \lhd_{t_0} F$ if the following conditions hold:

- $\{T_1, \dots, T_k\}$ is a partition of $V(F)$. Consequently, for any $i \in [k]$, $T_i \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$, by restriction.
- For any $i \in [k]$, $T_i \in \mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$.

If $\{T_1, \dots, T_k\} \lhd_{t_0} F$, we denote by $F/\{T_1, \dots, T_k\}$ the forest obtained by contracting T_i to a single vertex for any $i \in [k]$, decorating this vertex by T_i, and forgetting the type of the remaining edges. Then $F/\{T_1, \dots, T_k\}$ is a $\mathcal{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$-decorated forest.

Proposition 5.9. *Let $\lambda \in \mathbb{K}^{\mathcal{T}}$, $t_0 \in \mathcal{T}$. Let us consider the map*

$$\Psi^*_{t_0} : \begin{cases} \mathcal{H}_{\mathcal{D},\mathcal{T}} \longrightarrow \mathcal{H}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}, \\ F \in \mathbb{F}_{\mathcal{D},\mathcal{T}} \longmapsto \displaystyle\sum_{\{T_1,\dots,T_k\} \lhd_{t_0} F} \left(\prod_{e \in E(F) \setminus \sqcup E(T_i)} \lambda_{\mathrm{type}(e)} \right) F/\{T_1, \dots, T_k\}. \end{cases}$$

Then $\Psi^*_{t_0}$ is a Hopf algebra morphism from $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ to $\mathcal{H}^{\mathrm{CK}}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}$. It is an isomorphism if, and only if, $\lambda_{t_0} \neq 0$.

Proof. First case. We first assume that $\mathcal{D}$ and $\mathcal{T}$ are finite. In this case, $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ is the graded dual of $\mathcal{H}^{\mathrm{GL}_\lambda}_{\mathcal{D},\mathcal{T}}$, with the Hopf pairing $\langle -, - \rangle$; grading $\mathcal{H}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}$ by the number of vertices of the decorations, $\mathcal{H}^{\mathrm{CK}}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}$ is the graded dual of $\mathcal{H}^{\mathrm{GL}}_{\mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}}$. Moreover, $\Psi^*_{t_0}$ is the transpose of Ψ_{t_0} of Proposition 4.9, so is a Hopf algebra morphism. If $\lambda_{t_0} \neq 0$, Ψ_{t_0} is an isomorphism, so $\Psi^*_{t_0}$ also is.

General case. Let $x, y \in \mathcal{H}_{\mathcal{D},\mathcal{T}}$. There exist finite $\mathcal{D}', \mathcal{T}'$, such that $x, y \in \mathcal{H}_{\mathcal{D}',\mathcal{T}'}$; we can assume that $t_0 \in \mathcal{T}'$. We denote by $\lambda' = \lambda_{|\mathcal{T}'}$. Then, by the preceding case, denoting by Ψ'_{t_0} the restriction of $\Psi^*_{t_0}$ to $\mathcal{H}_{\mathcal{D}',\mathcal{T}'}$,

$$\Psi^*_{t_0}(xy) = \Psi'_{t_0}(xy) = \Psi'_{t_0}(x)\Psi'_{t_0}(y) = \Psi^*_{t_0}(x)\Psi^*_{t_0}(y),$$

$$\Delta^{\mathrm{CK}_\lambda} \circ \Psi^*_{t_0}(x) = \Delta^{\mathrm{CK}_{\lambda'}} \circ \Psi'_{t_0}(x) = (\Psi'_{t_0} \otimes \Psi'_{t_0}) \circ \Delta^{\mathrm{CK}_{\lambda'}}(x) = (\Psi^*_{t_0} \otimes \Psi^*_{t_0}) \circ \Delta^{\mathrm{CK}_\lambda}(x),$$

so Ψ is a Hopf algebra morphism.

Let us assume that $\lambda_{t_0} \neq 0$. If $\Psi^*_{t_0}(x) = 0$, then $\Psi'_{t_0}(x) = 0$. As $a'_{t_0} \neq 0$, by the first case, $x = 0$, so $\Psi^*_{t_0}$ is injective. Moreover, there exists $z \in \mathcal{H}_{\mathcal{D}',\mathcal{T}'}$, such that $\Psi'_{t_0}(z) = y$; so $\Psi^*_{t_0}(z) = y$, and $\Psi^*_{t_0}$ is surjective.

Let us assume that $\lambda_{t_0} = 0$. Let T be a tree with two vertices, such that its unique edge is of type t_0. As $T \notin \mathbb{T}^{(t_0)}_{\mathcal{D},\mathcal{T}}$, $\Phi_{t_0}(T)$ has a unique term, given by the partition $X = \{\{x_1\}, \{x_2\}\}$, where x_1 and x_2 are the vertices of T. Hence,

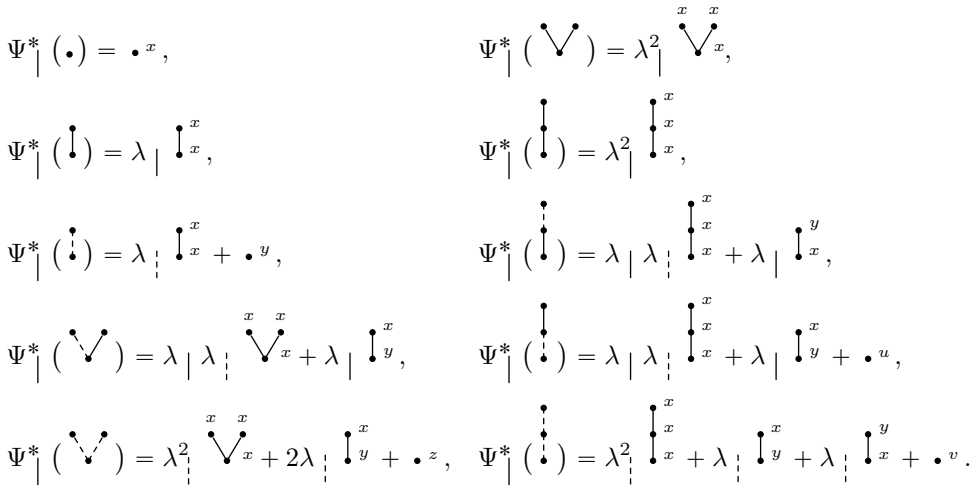
$$\Psi^*_{t_0}(T) = \lambda_{t_0} T' = 0,$$

so $\Psi^*_{t_0}$ is not injective. □

Example 5.10. Here, $\mathcal{T}$ contains two elements, $|$ and $\vdots$. In order to simplify, we omit the decorations of vertices. We put

$x =$, $y =$, $z =$, $u =$, $v =$.

Applying $\Psi^*_{|}$:



Remark 5.11. Although it is not indicated, Ψ_{t_0} and $\Psi^*_{t_0}$ depend on λ.

5.5. **Bialgebras in cointeraction.** By [12], for any $\lambda \in \mathbb{K}^{(\mathcal{T})}$, the operad morphism $\theta_\lambda \colon \mathbf{PreLie} \longrightarrow \mathcal{P}_\mathcal{T}$, which send $\bullet$ to $\bullet_\lambda$, where **PreLie** is the operad of pre-Lie algebras, induces a pair of cointeracting bialgebras for any finite set $\mathcal{D}$. By construction, the first bialgebra of the pair is $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$. Let us describe the second one.

Definition 5.12. Let $F \in \mathbb{F}_{\mathcal{T},\mathcal{D}}$. We shall say that $\{T_1, \dots, T_k\} \lhd F$ if

(1) $\{T_1, \dots, T_k\}$ is a partition of $V(F)$. Consequently, for any $i \in [k]$, $T_i \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$, by restriction.
(2) For any $i \in [k]$, $T_i \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$.

If $\{T_1, \dots, T_k\} \lhd F$ and $\mathrm{dec} \colon [k] \longrightarrow \mathcal{D}$, we denote by $(F/\{T_1, \dots, T_k\}, \mathrm{dec})$ the forest obtained by contracting T_i to a single vertex, and decorating this vertex by $\mathrm{dec}(i)$, for all $i \in [k]$. This is an element of $\mathbb{F}_{\mathcal{D},\mathcal{T}}$.

Proposition 5.13. *If $\mathcal{D}$ is finite, $\mathcal{H}'_{\mathcal{D},\mathcal{T}}$ is the free commutative algebra generated by pairs (T, d), where $T \in \mathbb{T}_{\mathcal{T},\mathcal{D}}$ and $d \in \mathcal{D}$. The coproduct is given, for any $F \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$, $d \in \mathcal{D}$, by*

$$\delta(F, d) = \sum_{\{T_1,\dots,T_k\} \lhd F} \sum_{\mathrm{dec} \colon [k] \longrightarrow \mathcal{D}} ((F/\{T_1, \dots, T_k\}, \mathrm{dec}), d) \otimes (T_1, \mathrm{dec}(1)) \cdots (T_k, \mathrm{dec}(k)).$$

Then $(\mathcal{H}'_{\mathcal{D},\mathcal{T}}, m, \delta)$ is a bialgebra, and $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ is a coalgebra in the category of $\mathcal{H}'_{\mathcal{D},\mathcal{T}}$-comodules via the coaction given, for any $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$, by

$$\overline{\delta}(T) = \sum_{\{T_1,\dots,T_k\} \lhd T} \sum_{\mathrm{dec} \colon [k] \longrightarrow \mathcal{D}} ((T/\{T_1, \dots, T_k\}, \mathrm{dec}) \otimes (T_1, \mathrm{dec}(1)) \cdots (T_k, \mathrm{dec}(k)).$$

Corollary 5.14. *Let us assume that $\mathcal{D}$ is given a semigroup law denoted by $+$. If $F \in \mathbb{F}_{\mathcal{T},\mathcal{D}}$, and $\{T_1, \dots, T_k\} \lhd F$, then naturally $T_i \in \mathbb{T}_{\mathcal{T},\mathcal{D}}$ for any i and the $\mathcal{T}$-typed forest $F/\{T_1, \dots, T_k\}$ is given a $\mathcal{D}$-decoration, decorating the vertex obtained in the contradiction of T_i by the sum of the decorations of the vertices of T_i. Then $\mathcal{H}_{\mathcal{D},\mathcal{T}}$ is given a second coproduct δ such that for any $F \in \mathbb{F}_{\mathcal{D},\mathcal{T}}$,*

$$\delta(F) = \sum_{\{T_1,\dots,T_k\} \lhd F} F/\{T_1, \dots, T_k\} \otimes T_1 \cdots T_k.$$

Then $(\mathcal{H}_{\mathcal{D},\mathcal{T}}, m, \delta)$ is a bialgebra and $\mathcal{H}^{\mathrm{CK}_\lambda}_{\mathcal{D},\mathcal{T}}$ is a coalgebra in the category of $\mathcal{H}_{\mathcal{D},\mathcal{T}}$-comodules via the coaction δ.

Proof. We denote by I the ideal of $\mathcal{H}'_{\mathcal{D},\mathcal{T}}$ generated by pairs (T,d) such that $T \in \mathbb{T}_{\mathcal{D},\mathcal{T}}$ and $d \in \mathcal{D}$, with

$$d \neq \sum_{v \in V(T)} \mathrm{dec}(v).$$

The quotient $\mathcal{H}'_{\mathcal{D},\mathcal{T}}/I$ is identified with $\mathcal{H}_{\mathcal{D},\mathcal{T}}$, trough the surjective algebra morphism

$$\varpi\colon \begin{cases} \mathcal{H}'_{\mathcal{D},\mathcal{T}} \longrightarrow \mathcal{H}_{\mathcal{D},\mathcal{T}}, \\ (F,d) \in \mathbb{F}_{\mathcal{D},\mathcal{T}} \times \mathcal{D} \longmapsto \begin{cases} F & \text{if } d = \sum_{v \in V(F)} \mathrm{dec}(v), \\ 0 & \text{otherwise.} \end{cases} \end{cases}$$

Let us prove that I is a coideal. Let $T \in \mathbb{T}_{\mathcal{T},\mathcal{D}}$, $d \in \mathcal{D}$, $\{T_1,\dots,T_k\} \lhd F$, $\mathrm{dec}\colon [k] \longrightarrow \mathcal{D}$ such that $((T/\{\mathcal{T}_1,\dots,T_k\},\mathrm{dec}),d) \notin I$ and for any i, $(T_i,\mathrm{dec}(i)) \notin I$. Then

$$\forall i \in [k], \qquad \sum_{v \in V(T_i)} \mathrm{dec}(v) = \mathrm{dec}(i), \qquad \sum_{i=1}^{k} \mathrm{dec}(i) = d.$$

Hence,

$$\sum_{v \in V(T)} \mathrm{dec}(v) = \sum_{i=1}^{k} \sum_{v \in V(T_i)} \mathrm{dec}(v) = \sum_{i=1}^{k} \mathrm{dec}(i) = d,$$

so $(T,d) \notin I$. Consequently, if $T \in I$, then $((T/\{\mathcal{T}_1,\dots,T_k\},\mathrm{dec}),d) \in I$ or at least one of the $(T_i,\mathrm{dec}(i))$ belongs to I. Hence,

$$\delta(I) \subseteq I \otimes \mathcal{H}'_{\mathcal{D},\mathcal{T}} + \mathcal{H}'_{\mathcal{D},\mathcal{T}} \otimes I.$$

So I is a coideal. The coproduct induced on $\mathcal{H}_{\mathcal{D},\mathcal{T}}$ by the morphism ϖ is precisely the one given in the setting of this corollary. □

In particular, if $\mathcal{D}$ is reduced to a single element, denoted by $*$, if we give it its unique semigroup structure ($* + * = *$), We obtain again the result of [7].

5.6. **The Bruned–Hairer–Zambotti construction.** We now consider the coproducts on typed trees in [3, Theorem 2.2.16], the first one with $\overline{\mathfrak{A}}(F) = \mathfrak{A}(F)$ and the second one with $\overline{\mathfrak{A}}(F) = \mathfrak{A}^+(F)$ of [3, Definition 2.4.1]. By definition [3, Definition 2.26] of admissible subtrees, according to the notations we choose in this paper:

- Let $\mathcal{L}$ be a finite set of types. The considered trees are $\mathcal{L}$-typed and the leaves are $\mathcal{L}$-decorated. Considering that the internal vertices of such a tree are in fact decorated by an element $0 \notin \mathcal{L}$, these trees form a subset of $\mathbb{T}_{\mathcal{L}\sqcup\{0\},\mathcal{L}}$ which we denote by $\mathbb{T}'_{\mathcal{L}}$.
- The first coproduct Δ_+ is given on any tree $T \in \mathbb{T}'_{\mathcal{L}}$ by

$$\Delta_+(T) = \sum_{c \in \mathrm{Adm}'(T)} R^c(T) \otimes P^c(T),$$

  where $\mathrm{Adm}'(T)$ is the set of admissible cuts of T such that the set of leaves of $R^c(T)$ is a subset of leaves of T (note that automatically, the leaves of $P^c(T)$ are also leaves of T).
- The second coproduct is given on any tree $T \in \mathbb{T}'_{\mathcal{L}}$ by

$$\overline{\Delta}(T) = \sum_{\{T_1,\dots,T_k\} \lhd' T} T/\{T_1,\dots,T_k\} \otimes T_1 \cdots T_k,$$

  where the sum runs over all $\{T_1,\dots,T_k\} \lhd T$ such that the leaves of $T/\{T_1,\dots,T_k\}$ are leaves of T.

We denote by $\mathcal{H}'_{\mathcal{L}}$ the subalgebra of $\mathcal{H}_{\mathcal{L}\sqcup\{0\},\mathcal{L}}$ generated by $\mathbb{T}'_{\mathcal{L}}$. Then $(\mathcal{H}'_{\mathcal{L}}, m, \Delta_+)$ and $(\mathcal{H}'_{\mathcal{L}}, m, \delta)$ are bialgebras.

Let us give to $\mathcal{L}$ any product $\times$ making it a commutative associative semigroup. We extend this structure to $\mathcal{L} \sqcup \{0\}$ by

$$\forall t \in \mathcal{L} \sqcup \{0\}, \qquad 0 \times t = t \times 0 = 0.$$

We take $\lambda_t = 1$ for any $t \in \mathcal{L} \sqcup \{0\}$, and obtain with this data two coproducts Δ and δ on $\mathcal{H}_{\mathcal{L}\sqcup\{0\},\mathcal{L}}$, the first one given by admissible cuts and the second one by contractions of subtrees. The subalgebra $\mathcal{H}''_{\mathcal{L}}$ of $\mathcal{H}_{\mathcal{L}\sqcup\{0\},\mathcal{L}}$ generated by trees such that any internal vertex is decorated by 0 (note that the leaves of such a tree are decorated by $L \sqcup \{0\}$) is a subbialgebra for both coproducts. We denote by I the ideal of $\mathcal{H}''_{\mathcal{L}}$ generated by trees with at least one leaf decorated by 0. Then it is a coideal for both coproducts, so the quotient algebra $\mathcal{H}''_{\mathcal{L}}/I$ inherits two coproducts, still denoted by Δ and δ. This algebra is trivially identified with the algebra generated by $\mathbb{T}'_{\mathcal{L}}$, that is to say with $\mathcal{H}'_{\mathcal{L}}$. By construction of the different coproducts, this identification is an isomorphism from the Hopf algebra $(\mathcal{H}''_{\mathcal{L}}/I, \Delta)$ to $(\mathcal{H}'_{\mathcal{L}}/I, \Delta_+)$ and from the Hopf algebra $(\mathcal{H}''_{\mathcal{L}}/I, \delta)$ to $(\mathcal{H}'_{\mathcal{L}}/I, \overline{\Delta})$. In other words, the Bruned–Hairer–Zambotti construction of cointeracting bialgebras on typed trees is a subquotient of the construction presented here.

6. Appendix

6.1. **Values of $t_{D,T}(k)$.** For $D = 1$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	1	1	2	4	9	20	48	115
2	1	2	7	26	107	458	2058	9498
3	1	3	15	82	495	3144	20875	142773
4	1	4	26	188	1499	12628	111064	1006840
5	1	5	40	360	3570	37476	410490	4635330
6	1	6	57	614	7284	91566	1200705	16232820
7	1	7	77	966	13342	195384	2984142	46990952
8	1	8	100	1432	22570	377320	6578116	118238600
9	1	9	126	2028	35919	674964	13225632	267188229
10	1	10	155	2770	54465	1136402	24723000	554540590

For $D = 2$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	2	4	14	52	214	916	4116	18996
2	2	8	52	376	2998	25256	222128	2013680
3	2	12	114	1228	14568	183132	2401410	32465640
4	2	16	200	2864	45140	754640	13156232	236477200
5	2	20	310	5540	108930	2272804	49446000	1109081180
6	2	24	444	9512	224154	5606520	146204792	3930863232
7	2	28	602	15036	413028	12043500	366122190	11475005616
8	2	32	784	22368	701768	23373216	811575408	29052861280
9	2	36	990	31764	1120590	41969844	1638712716	65965167108
10	2	40	1220	43480	1703710	70875208	3073688160	137426005200

For $D = 3$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	3	9	45	246	1485	9432	62625	428319
2	3	18	171	1842	21852	274698	3602115	48698460
3	3	27	378	6084	107757	2024892	39676896	801564687
4	3	36	666	14268	336231	8409780	219307188	5896294848
5	3	45	1035	27690	814680	25444584	828506340	27812997990
6	3	54	1485	47646	1680885	62954766	2458069074	98947750662
7	3	63	2016	75432	3103002	135520812	6170116638	289616448690
8	3	72	2628	112344	5279562	263423016	13701398868	734709311208
9	3	81	3321	159678	8439471	473586264	27703353159	1670715963729
10	3	90	4095	218730	12842010	800524818	52018920345	3484841027040

For $D = 4$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	4	16	104	752	5996	50512	444256	4027360
2	4	32	400	5728	90280	1509280	26312464	472954400
3	4	48	888	19024	448308	11213040	292409584	7861726464
4	4	64	1568	44736	1403536	46746432	1623150816	58105722560
5	4	80	2440	86960	3407420	141750416	6147376320	274852010400
6	4	96	3504	149792	7039416	351230688	18268531824	979612414944
7	4	112	4760	237328	13006980	756866096	45910215120	2871018269632
8	4	128	6208	353664	22145568	1472317056	102037088448	7290356719488
9	4	144	7848	502896	35418636	2648533968	206451156768	16590568445280
10	4	160	9680	689120	53917640	4479065632	387863411920	34625886677920

we find the following sequences of the OEIS [24]:

$T\backslash D$	1	2	3	4
1	A0081	A038055	A038059	A136793
2	A00151	A136794		
3	A006964			
4	A052763			
5	A052788			
6	A246235			
7	A246236			
8	A246237			
9	A246238			
10	A246239			

6.2. **Values of $t'_{D,T}(k)$.** For $D = 1$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	1	0	0	0	0	0	0	0
2	1	1	3	10	39	160	702	3177
3	1	2	9	46	268	1660	10845	73270
4	1	3	18	124	963	7968	69236	621999
5	1	4	30	260	2525	26136	283528	3178696
6	1	5	45	470	5480	68096	885805	11904160
7	1	6	63	770	10479	151956	2304974	36110880
8	1	7	84	1176	18298	303296	5255964	94051770
9	1	8	108	1704	29838	556464	10845732	218239560
10	1	9	135	2370	46125	955872	20696076	462558987

For $D = 2$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	2	0	0	0	0	0	0	0
2	2	4	22	144	1090	8864	76162	678532
3	2	8	68	688	7886	96896	1250780	16713504
4	2	12	138	1888	29004	476736	8213588	146342376
5	2	16	232	4000	77060	1586304	34185344	761389360
6	2	20	350	7280	168670	4171744	107932710	2884827980
7	2	24	492	11984	324450	9370368	282934428	8822987856
8	2	28	658	18368	569016	18793600	648698792	23119514576
9	2	32	848	26688	930984	34609920	1344232416	53898191520
10	2	36	1062	37200	1442970	59627808	2573660298	114661732500

For $D = 3$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	3	0	0	0	0	0	0	0
2	3	9	72	705	7947	96588	1237728	16450389
3	3	18	225	3408	58347	1072224	20685195	413084610
4	3	27	459	9405	216081	5315112	136987407	3650993163
5	3	36	774	19992	576405	17763984	572991726	19100718828
6	3	45	1170	36465	1264950	46852884	1815034140	72635168880
7	3	54	1647	60120	2437722	105455952	4768982442	222723271080
8	3	63	2205	92253	4281102	211832208	10953036318	584744300226
9	3	72	2844	134160	7011846	390570336	22727284344	1365242802048
10	3	81	3564	187137	10877085	673533468	43560017892	2907844041231

For $D = 4$,

$T\backslash k$	1	2	3	4	5	6	7	8
1	4	0	0	0	0	0	0	0
2	4	16	168	2192	32844	531200	9051376	159962784
3	4	32	528	10656	242792	5939968	152518064	4053650976
4	4	48	1080	29488	902100	29551104	1014147872	35989518528
5	4	64	1824	62784	2411024	98976256	4252211232	188790415552
6	4	80	2760	114640	5297820	261422336	13491005840	719200139360
7	4	96	3888	189152	10218744	588999936	35487727184	2208096700896
8	4	112	5208	290416	17958052	1184031744	81574704960	5802692175744
9	4	128	6720	422528	29428000	2184360960	169377005376	13557899008896
10	4	144	8424	589584	45668844	3768659712	324805399344	28894042642464

References

[1] Bergeron F., Labelle G., Leroux P., Combinatorial species and tree-like structures, *Encyclopedia of Mathematics and its Applications*, Vol. 67, Cambridge University Press, Cambridge, 1998.

[2] Bremner M.R., Dotsenko V., Algebraic operads. An algorithmic companion, CRC Press, Boca Raton, FL, 2016.

[3] Bruned Y., Singular KPZ type equations, avalaible at `https://tel.archives-ouvertes.fr/tel-01306427`.

[4] Bruned Y., Chandra A., Chevyrev I., Hairer M., Renormalising SPDEs in regularity structures, *J. Eur. Math. Soc. (JEMS)* **23** (2021), 869–947, arXiv:1711.10239.

[5] Bruned Y., Hairer M., Zambotti L., Algebraic renormalisation of regularity structures, *Invent. Math.* **215** (2019), 1039–1156, arXiv:1610.08468.

[6] Bruned Y., Manchon D., Algebraic deformation for SPDEs, arXiv:2011.05907.
[7] Calaque D., Ebrahimi-Fard K., Manchon D., Two interacting Hopf algebras of trees: a Hopf-algebraic approach to composition and substitution of B-series, *Adv. in Appl. Math.* **47** (2011), 282–308, arXiv:0806.2238.
[8] Chapoton F., Un endofoncteur de la catégorie des opérades, in Dialgebras and Related Operads, *Lecture Notes in Math.*, Vol. 1763, Springer, Berlin, 2001, 105–110.
[9] Chapoton F., Livernet M., Pre-Lie algebras and the rooted trees operad, *Int. Math. Res. Not.* **2001** (2001), 395–408, arXiv:math.QA/0002069.
[10] Connes A., Kreimer D., Hopf algebras, renormalization and noncommutative geometry, *Comm. Math. Phys.* **199** (1998), 203–242, arXiv:hep-th/9808042.
[11] Dotsenko V., Griffin J., Cacti and filtered distributive laws, *Algebr. Geom. Topol.* **14** (2014), 3185–3225, arXiv:1109.5345.
[12] Foissy L., Algebraic structures associated to operads, arXiv:1702.05344.
[13] Foissy L., Generalized prelie and permutative algebras, arXiv:2104.00909.
[14] Grossman R., Larson R.G., Hopf-algebraic structure of combinatorial objects and differential operators, *Israel J. Math.* **72** (1990), 109–117.
[15] Hoffman M.E., Combinatorics of rooted trees and Hopf algebras, *Trans. Amer. Math. Soc.* **355** (2003), 3795–3811, arXiv:math.CO/0201253.
[16] Livernet M., A rigidity theorem for pre-Lie algebras, *J. Pure Appl. Algebra* **207** (2006), 1–18, arXiv:math.QA/0504296.
[17] Loday J.-L., Vallette B., Algebraic operads, *Grundlehren der Mathematischen Wissenschaften*, Vol. 346, Springer, Heidelberg, 2012.
[18] Manchon D., Zhang Y., Free pre-Lie family algebras, arXiv:2003.00917.
[19] Markl M., Shnider S., Stasheff J., Operads in algebra, topology and physics, *Mathematical Surveys and Monographs*, Vol. 96, Amer. Math. Soc., Providence, RI, 2002.
[20] Mathar R.J., Topologically distinct sets of non-intersecting circles in the plane, arXiv:1603.00077.
[21] Oudom J.-M., Guin D., Sur l'algèbre enveloppante d'une algèbre pré-Lie, *C. R. Math. Acad. Sci. Paris* **340** (2005), 331–336.
[22] Oudom J.-M., Guin D., On the Lie enveloping algebra of a pre-Lie algebra, *J. K-Theory* **2** (2008), 147–167, arXiv:math.QA/0404457.
[23] Panaite F., Relating the Connes–Kreimer and Grossman–Larson Hopf algebras built on rooted trees, *Lett. Math. Phys.* **51** (2000), 211–219, arXiv:math.QA/0003074.
[24] Sloane N.J.A., The on-line encyclopedia of integer sequences, `http://oeis.org/`.
[25] Zhang Y., Gao X., Guo L., Matching Rota–Baxter algebras, matching dendriform algebras and matching pre-Lie algebras, *J. Algebra* **552** (2020), 134–170, arXiv:1909.10577.